

# **ESAT CAMP MOCK D**

Full sitting

EC-FULL-D

**MATHEMATICS 1**

**40 MINUTES**

**MATHEMATICS 2**

**40 MINUTES**

**PHYSICS**

**40 MINUTES**

**CHEMISTRY**

**40 MINUTES**

**BIOLOGY**

**40 MINUTES**

## **INSTRUCTIONS TO CANDIDATES**

Please read these instructions carefully, but **do not open this question paper until you are ready to begin**.

This paper contains **5 modules** (Mathematics 1, Mathematics 2, Physics, Chemistry, Biology). Each module has **27 multiple-choice questions** and is designed for **40 minutes** (about **200 minutes** in total if you sit every module).

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

**Please wait until you are ready before turning this page.**

*Original ESAT CAMP practice material. Not an official UAT-UK or Pearson paper.*

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## **Mathematics 1**

- 1 A graphic designer creates a square logo of side length  $L$ .  
The logo is first reduced in size so that its side length decreases by 25%.  
Realising the logo is too small, the designer attempts to restore it to its original size by increasing the side length of the reduced logo by 25%.  
The area of this final square logo is  $31 \text{ cm}^2$  smaller than the area of the original square logo.  
What is the perimeter of the original square logo?
- A 32 cm  
B 48 cm  
C 60 cm  
D 64 cm  
E 256 cm
- 2 A set of numbers is  $S = \{1, 2, 3, 4, 5\}$ . A number is chosen at random from the set  $S$  three times, with replacement.  
What is the probability that the product of the three chosen numbers is an even number?
- A  $\frac{8}{125}$   
B  $\frac{27}{125}$   
C  $\frac{54}{125}$   
D  $\frac{98}{125}$   
E  $\frac{117}{125}$   
F  $\frac{1}{2}$
- 3 The value of an asset increases by 60%. The new value then decreases by 25%. The final value of the asset is 1080.  
What was the initial value of the asset?
- A 540  
B 800  
C 864  
D 900  
E 1296

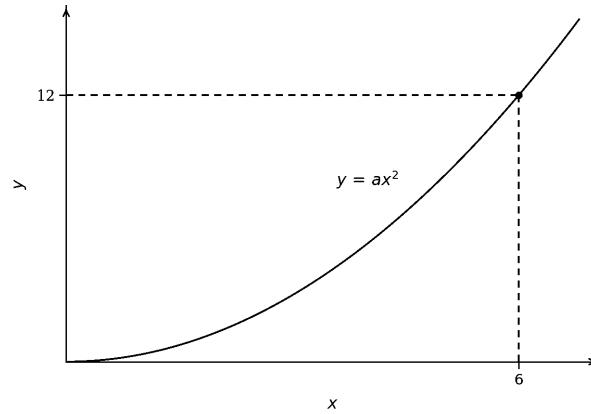
- 4 A car travels a certain distance at 30 km/h. It then travels twice that distance at 90 km/h. What is the average speed of the car for the whole journey?
- A 40 km/h
  - B 45 km/h
  - C 54 km/h
  - D 60 km/h
  - E 70 km/h
- 5 A quantity  $x$  is increased by 150% to give quantity  $y$ . Quantity  $z$  is  $q\%$  less than  $y$ . Quantity  $z$  is also 50% less than  $x$ . What is the value of  $q$ ?
- A 20
  - B 40
  - C 80
  - D 100
  - E 200

- 6 The diagram shows the graph of  $y = ax^2$  for  $x \geq 0$ , where  $a$  is a constant. The curve passes through the point  $(6, 12)$ .

The variable  $x$  is inversely proportional to  $t$ .

When  $t = 4$ ,  $x = 3$ .

Which of the following gives an expression for  $y$  in terms of  $t$ ?



- A  $y = \frac{4}{t}$
- B  $y = \frac{4}{t^2}$
- C  $y = \frac{3}{16t^2}$
- D  $y = \frac{48}{t^2}$
- E  $y = \frac{288}{t^2}$
- F  $y = \frac{432}{t^2}$
- G  $y = \frac{3t^2}{16}$
- H  $y = 48t^2$
- 7 A composite cylindrical rod has length  $L$  and total radius  $2r$ . It consists of a solid inner core of radius  $r$  and a surrounding outer sleeve. The core is made of a material with density  $3\rho$  and the sleeve is made of a material with density  $\rho$ . Which expression gives the total mass of the rod?

- A  $4\pi r^2 L \rho$
- B  $6\pi r^2 L \rho$
- C  $7\pi r^2 L \rho$
- D  $8\pi r^2 L \rho$
- E  $10\pi r^2 L \rho$
- F  $12\pi r^2 L \rho$

- 8 A digital file of size 120 Megabytes is to be transmitted over a network. The transmission protocol requires that for every 15 parts of data, 1 part of overhead is added.

The network has a constant transmission speed of 32 Megabits per second.

Assuming 1 Megabyte =  $10^6$  bytes, and 1 byte = 8 bits, what is the total time required to transmit the file and its overhead?

- A 0.5 seconds
- B 2 seconds
- C 4 seconds
- D 30 seconds
- E 32 seconds
- F 480 seconds

- 9 Pipe P can fill a tank in 10 hours. Pipe Q can fill the same tank in  $T$  hours.

Pipe P works alone for 2 hours to begin filling the empty tank. Then pipe Q is also opened, and working together, they take a further 4 hours to fill the tank.

What is the value of  $T$ ?

- A 5
- B 7.5
- C 8
- D 10
- E 15
- F 20

- 10 Two quadratic equations of the form  $x^2 + 2bx + c = 0$  are formed.

For each equation, the coefficients  $b$  and  $c$  are chosen independently and uniformly at random from the set  $\{1, 2, 3\}$ .

What is the probability that at least one of the two equations has real roots?

- A  $\frac{4}{81}$
- B  $\frac{28}{81}$
- C  $\frac{49}{81}$
- D  $\frac{7}{9}$
- E  $\frac{8}{9}$
- F  $\frac{77}{81}$

- 11 The finite region  $R$  is bounded by the four lines with equations:

$$3x + 4y = 22$$

$$3x - 4y = 14$$

$$3x + 4y = -2$$

$$3x - 4y = -10$$

What is the area of  $R$ ?

- A 12
  - B 20
  - C 24
  - D 25
  - E 48
- 12 The positive integers  $x, y, z$  have no common factor strictly greater than 1. They satisfy the relationship:

$$3x : 4y : 5z = 2 : 3 : 4$$

Given that  $k$  is a constant such that  $kx + y - z = 77$ , what is the value of  $k$ ?

- A 2
  - B 26
  - C 27
  - D 38
  - E 39
  - F 40
- 13 Seven distinct items are arranged in a row. How many arrangements are possible in which no two of three specific items are adjacent?
- A 240
  - B 576
  - C 720
  - D 1440
  - E 4320

- 14** A curve is given by the equation  $y = (x - k)(x^2 - 4)$ , where  $k$  is a real constant such that  $k > 1$ .  
A dataset consists of the  $y$ -intercept and all the  $x$ -intercepts of the curve.

The mean of this dataset is equal to its median.

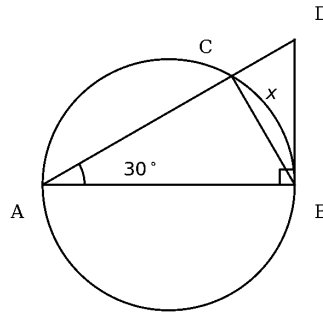
What is the value of  $k$ ?

- A**  $\frac{2}{9}$
- B**  $\frac{4}{7}$
- C**  $\frac{4}{3}$
- D**  $\frac{8}{5}$
- E**  $\frac{8}{3}$
- F** 6

- 15 The diagram shows a circle with diameter  $AB$ .

The straight line  $AD$  passes through point  $C$  on the circumference of the circle, and the line  $BD$  is tangent to the circle at  $B$ .

Given that  $\angle CAB = 30^\circ$  and the length of  $BC$  is  $x$ , which of the following is a correct expression for the length of  $CD$  in terms of  $x$ ?



[diagram not to scale]

- A  $\frac{1}{3}x$   
 B  $\frac{1}{2}x$   
 C  $\frac{\sqrt{3}}{6}x$   
 D  $\frac{\sqrt{3}}{3}x$   
 E  $\frac{\sqrt{3}}{2}x$   
 F  $\frac{2\sqrt{3}}{3}x$   
 G  $\sqrt{3}x$   
 H  $2\sqrt{3}x$
- 16 Two cones have the same volume. The height of the second cone is twice the height of the first cone.

If the base radius of the first cone is  $r$ , what is the base radius of the second cone?

- A  $\frac{r}{\sqrt{2}}$   
 B  $\frac{r}{2}$   
 C  $\frac{r}{4}$   
 D  $r\sqrt{2}$   
 E  $2r$   
 F  $4r$

- 17** The curve  $C$  has equation  $y = 6x - x^2$ .

The point  $M$  is the maximum point of  $C$ .

The points  $P$  and  $Q$  have position vectors  $\vec{OP}$  and  $\vec{OQ}$  such that:

$$3\vec{OM} = \vec{OP} + 2\vec{OQ}$$

where  $O$  is the origin.

Given that  $P$  has coordinates  $(7, 3)$ , what is the distance from  $Q$  to the point where  $C$  intersects the positive  $x$ -axis?

- A** 13
- B**  $\sqrt{13}$
- C**  $\sqrt{73}$
- D**  $\sqrt{145}$
- E**  $\sqrt{229}$
- F**  $4\sqrt{37}$

- 18** Liquid X has a density of 4.

Liquid A has a density of 5.

Liquid B has a density of 2.

A mixture, Y, is created by combining liquids A and B. A new blend is then formed by mixing liquid X and mixture Y.

In this final blend, the ratio of the volume of X to the volume of Y is 1 : 3.

The ratio of the mass of X to the mass of Y is 2 : 5.

What is the ratio of the volume of liquid A to the volume of liquid B in mixture Y?

- A** 1 : 1
- B** 2 : 1
- C** 4 : 5
- D** 5 : 4
- E** 14 : 1
- F** 4 : 25

- 19** A fair six-sided die is rolled.  
If the number rolled is a multiple of 3, a fair coin is tossed 3 times.  
Otherwise, the coin is tossed 2 times.  
What is the probability of obtaining exactly 2 heads?

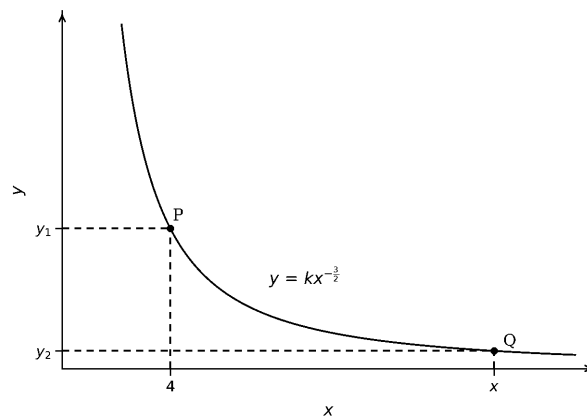
- A**  $\frac{5}{24}$   
**B**  $\frac{13}{48}$   
**C**  $\frac{7}{24}$   
**D**  $\frac{5}{16}$   
**E**  $\frac{1}{3}$   
**F**  $\frac{3}{8}$

- 20** The diagram shows part of the curve with equation  $y = kx^{-\frac{3}{2}}$ , where  $k$  is a positive constant and  $x > 0$ .

The points  $P(4, y_1)$  and  $Q(x, y_2)$  lie on the curve.

The value of  $y_2$  is 87.5% less than the value of  $y_1$ .

What is the value of  $x$ ?



- A** 8  
**B**  $8\sqrt{2}$   
**C** 16  
**D** 32  
**E** 64

- 21** Three fair, six-sided dice are rolled. Let  $p$  be the probability that the highest value shown is 4. Let  $q$  be the probability that the lowest value shown is 2. What is the value of  $q - p$ ?

**A** 0  
**B**  $\frac{7}{72}$   
**C**  $\frac{1}{9}$   
**D**  $\frac{37}{216}$   
**E**  $\frac{61}{216}$

- 22** In a mathematical investigation, a shape is defined by the equation:

$$x^2 + y^2 - 4tx - 2ky + 4t^2 = 0$$

The parameter  $k$  is a fixed positive constant. The parameter  $t$  is a variable, and  $t > 0$ .

Which of the following statements about the shape is/are correct as  $t$  varies?

1. The area of the shape is constant.
2. The shape has exactly one point of intersection with the  $x$ -axis.
3. The distance of the center of the shape from the origin is a linear function of  $t$ .

**A** 1 only  
**B** 2 only  
**C** 1 and 2 only  
**D** 1 and 3 only  
**E** 2 and 3 only  
**F** 1, 2 and 3

- 23** A composite solid sphere of outer radius  $2r$  consists of a central spherical core of radius  $r$  and density  $\rho_c$ , surrounded by a shell of density  $\rho_s$ .

The mean density of the composite sphere is  $2\rho_s$ .

What is the ratio  $\frac{\rho_c}{\rho_s}$ ?

**A** 5  
**B** 7  
**C** 8  
**D** 9  
**E** 15

- 24** A data set  $S$  contains  $n$  distinct integers, where  $n \geq 2$ . Let  $\mu$  be its mean,  $m$  its median and  $r$  its range. A further integer data value  $k$  is appended to the data set; repeated values are allowed after this addition.

Which statements must be true?

1. If  $k > \mu$ , the new mean is greater than  $\mu$ .
2. If  $k > r$ , the new range is greater than  $r$ .
3. If  $k < m$ , the new median is less than  $m$ .

- A** 1 only  
**B** 1 and 2 only  
**C** 1 and 3 only  
**D** 2 and 3 only  
**E** 1, 2 and 3

- 25** For a regular polygon with  $n$  sides, the angle between a side extended and the adjacent side is  $(k - 20)^\circ$ , where  $k$  is a constant.

Which of the following expressions gives the value of  $n$ ?

- A**  $\frac{180}{k - 20}$   
**B**  $\frac{360}{k - 20}$   
**C**  $\frac{180}{200 - k}$   
**D**  $\frac{360}{200 - k}$

- 26** The variables  $x$  and  $y$  are such that the graph of  $y$  against  $x$  is a straight line through the origin. When  $x = 2$ ,  $y = 8$ .

The constants  $p$  and  $q$  are given by  $p = 5 \times 10^7$  and  $q = 2 \times 10^{-3}$ .

A quantity  $M$  is defined as:

$$M = \frac{\left(\frac{y}{q}\right)}{\left(\frac{x}{p}\right)}$$

What is the value of  $M$ ?

- A**  $1 \times 10^5$   
**B**  $6.25 \times 10^9$   
**C**  $2.5 \times 10^{10}$   
**D**  $1 \times 10^{11}$   
**E**  $1.6 \times 10^{-10}$   
**F**  $6.25 \times 10^3$

- 27 The real number  $x$  satisfies the equation:

$$\frac{x - \sqrt{3}}{x + \sqrt{3}} = 2 - \sqrt{3}$$

What is the value of  $x$ ?

- A -3
- B -1
- C 1
- D 2
- E 3

## Mathematics 2

- 1 How many distinct real solutions does the equation  $x^3 - 3x^2 + 5 = 0$  have?
- A 0
  - B 1
  - C 2
  - D 3
- 2 A circle has equation  $x^2 + y^2 = 50$ . A tangent to the circle at a point  $P$  is parallel to the line with equation  $7x + y = 1$ . The point  $P$  has positive  $x$  and  $y$  coordinates.
- What are the coordinates of  $P$ ?
- A (1, 7)
  - B (1, -7)
  - C (5, 5)
  - D (7, 1)
  - E (7, -1)
  - F (-7, -1)
- 3 A cubic polynomial  $p(x)$  has stationary points at  $x = 1$  and  $x = 3$ .
- What is the value of  $p(2)$ ?
- A -2
  - B -1
  - C 0
  - D 2
  - E 4
  - F Cannot be determined from the information given.

- 4 A box contains 4 red balls, 5 blue balls, and 3 green balls. All 12 balls are identical apart from their colour.

Balls are drawn at random from the box, one at a time, without replacement.

- If a red ball is drawn, it is not replaced and another ball is drawn.
- As soon as a blue ball or a green ball is drawn, the process stops.

What is the probability that at least two red balls are drawn and the process stops on a blue ball?

- A  $\frac{5}{132}$
- B  $\frac{1}{22}$
- C  $\frac{5}{88}$
- D  $\frac{5}{72}$
- E  $\frac{1}{11}$
- F  $\frac{5}{8}$

- 5 Find the value of the constant  $k$  for which the equation

$$2 \cos^2(x) - 3 \sin(x) = k$$

has exactly three distinct solutions in the interval  $0 \leq x < 2\pi$ .

- A  $k = -3$
- B  $k = -1$
- C  $k = 2$
- D  $k = \frac{25}{8}$
- E  $k = 3$
- F  $k = 5$

- 6 Variables  $x$  and  $y$  are related by the equation  $y = Ax^n$ , where  $A$  and  $n$  are constants. A graph of  $\log_2 y$  plotted against  $\log_2 x$  is a straight line with gradient 2 and intercept 4 on the vertical axis.

Which of the following expresses  $y$  in terms of  $x$ ?

- A  $y = 4x^2$
- B  $y = 16x^2$
- C  $y = 2x^4$
- D  $y = 4x^4$
- E  $y = 2^{2x+4}$

- 7 Consider the family of quadratic curves given by

$$y = x^2 - 2ax - 4a + 5$$

where  $a$  is a real constant.

What is the complete range of values of  $a$  for which the minimum point of the curve lies strictly in the second quadrant (that is, where  $x < 0$  and  $y > 0$ )?

- A** There are no such values of  $a$
- B**  $a < -5$
- C**  $-5 < a < 0$
- D**  $-5 < a < 1$
- E**  $0 < a < 1$
- F**  $a < -5$  or  $a > 1$
- G**  $a < -5$  or  $0 < a < 1$
- 8 The sum of the first 10 terms of a sequence is 200. Each term differs from the previous term by a constant amount. The sum of the next 10 terms is 600. What is the value of the first term?
- A** 0
- B** 1
- C** 2
- D** 4
- E** 11
- 9 A geometric series has first term  $a$  and common ratio  $r$ . All terms in the series are non-zero, and the sum to infinity of the series exists.

The sum of the first two terms is equal to  $\frac{5}{9}$  of the sum to infinity. The terms in the series do not all have the same sign.

What is the third term of the series, expressed as a fraction of the sum to infinity?

- A**  $\frac{4}{27}$
- B**  $\frac{19}{27}$
- C**  $\frac{20}{27}$
- D**  $\frac{4}{9}$
- E**  $\frac{35}{27}$
- F**  $\frac{-10}{9}$

- 10** Variables  $x$  and  $y$  are related by the equation  $y = kx^n$ , where  $k$  and  $n$  are constants. A graph of  $\log_{10} y$  against  $\log_{10} x$  is a straight line passing through the points  $(0, 2)$  and  $(4, 0)$ . Which of the following expresses  $y$  in terms of  $x$ ?
- A**  $y = 2 - \frac{x}{2}$
- B**  $y = \frac{2}{\sqrt{x}}$
- C**  $y = \frac{2}{x^2}$
- D**  $y = \frac{100}{\sqrt{x}}$
- E**  $y = \frac{100}{x^2}$
- 11** The function  $f(x) = x^3 - 3x^2 - 9x + 30$  is defined on the restricted domain  $-2 \leq x \leq 2$ . What is the range of  $f(x)$ ?
- A**  $3 \leq f(x) \leq 28$
- B**  $3 \leq f(x) \leq 35$
- C**  $8 \leq f(x) \leq 28$
- D**  $8 \leq f(x) \leq 35$
- E**  $-\infty < f(x) < \infty$
- 12** What is the coefficient of  $x^4$  in the expansion of  $(1 + 2x + 2x^2)^5$ ?
- A** 120
- B** 210
- C** 240
- D** 320
- E** 360

- 13 Consider the equation

$$\tan\left(\frac{4\pi}{x}\right) = \sqrt{3}$$

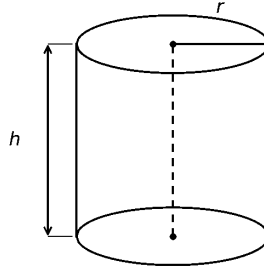
for  $x > 0$ .

The positive solutions are arranged in decreasing order,  $x_1 > x_2 > x_3 > \dots$

What is the value of  $x_3$ ?

- A  $\frac{6}{5}$
  - B  $\frac{12}{13}$
  - C  $\frac{12}{7}$
  - D  $\frac{24}{13}$
  - E 3
  - F 12
- 14 A sector of a circle has a fixed perimeter of 60.  
What is the radius of the sector that maximises its area?
- A  $30/\pi$
  - B  $60/(\pi + 2)$
  - C 15
  - D 20
  - E 30
- 15 The positive numbers  $\alpha$  and  $\beta$  satisfy the relation  $\alpha = 3\beta$ . A point  $P$  on the  $y$ -axis is a distance of 5 from the point  $(\alpha, 0)$  and a distance of 3 from the point  $(\beta, 0)$ .  
What is the value of the product  $\alpha\beta$ ?
- A 2
  - B 6
  - C 7
  - D 16
  - E 18
  - F 24

- 16** A solid right circular cylinder has base radius  $r$  and height  $h$ , as shown in the diagram. The base radius  $r$  is decreased by 20%, while the volume of the cylinder is increased by 28%. By what percentage does the height  $h$  increase?



*Diagram not to scale*

- A** 48%  
**B** 50%  
**C** 60%  
**D** 68%  
**E** 100%  
**F** 200%
- 17** What is the value of the sum

$$S = \sum_{n=2}^{49} \log_{10} \left( 1 - \frac{1}{n^2} \right)$$

- A**  $\log_{10} \left( \frac{25}{49} \right)$   
**B**  $\log_{10} \left( \frac{1}{2} \right)$   
**C**  $\log_{10} \left( \frac{50}{49} \right)$   
**D**  $\log_{10} \left( \frac{49}{50} \right)$   
**E**  $\log_{10} \left( \frac{24}{25} \right)$

- 18 Find the area of the finite region in the first quadrant bounded by the curve  $y = 2x^2$ , the line  $y = 32$ , and the  $y$ -axis.

- A  $\frac{128}{3}$   
 B 64  
 C  $\frac{256}{3}$   
 D 128  
 E  $\frac{320}{3}$   
 F  $\frac{512}{3}$

- 19  $ABCD$  is a trapezium with parallel sides  $AB$  and  $CD$ . The diagonals  $AC$  and  $BD$  intersect at point  $X$ , as shown in the diagram.

$AX = 1$  cm,  $AB = x$  cm,  $CX = (x + 2)$  cm, and  $CD = (4x + 1)$  cm.

What is the length, in cm, of  $CD$ ?

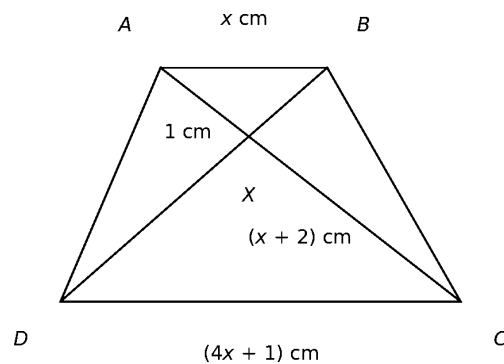


Diagram not to scale

- A  $3 + 2\sqrt{5}$   
 B  $4 + \sqrt{2}$   
 C  $4 + 4\sqrt{2}$   
 D  $5 + 4\sqrt{2}$   
 E  $5 + 4\sqrt{3}$   
 F  $5 + 8\sqrt{2}$

- 20** The polynomial  $(x + y)^{37}$  is expanded in ascending powers of  $x$ .  
How many of the terms in this expansion have a coefficient that is divisible by 37?
- A** 35  
**B** 36  
**C** 37  
**D** 38  
**E** 0
- 21** Which one of the following statements describes a mathematically impossible scenario for a polynomial  $p(x)$  with real coefficients?
- A** A quadratic polynomial  $p(x)$  has two distinct positive real roots and its derivative at  $x = 0$  is positive.  
**B** A cubic polynomial  $p(x)$  has a local maximum at  $x = 1$ , a local minimum at  $x = 4$ , and  $p(1) < p(4)$ .  
**C** A cubic polynomial  $p(x)$  has exactly one real root, and its derivative  $p'(x)$  has two distinct real roots.  
**D** A cubic polynomial  $p(x)$  has no stationary points and passes through the origin.  
**E** A quartic polynomial  $p(x)$  has exactly one stationary point.  
**F** A cubic polynomial  $p(x)$  has a local minimum at  $x = 1$  and a local maximum at  $x = 4$ .
- 22** A box contains 12 electronic components, of which 4 are defective and 8 are working. Components are drawn at random from the box, one at a time without replacement, and tested sequentially.  
Testing stops immediately once a second defective component has been identified.  
What is the probability that fewer working components than defective components have been tested when the process stops?
- A**  $\frac{1}{11}$   
**B**  $\frac{8}{55}$   
**C**  $\frac{9}{55}$   
**D**  $\frac{13}{55}$   
**E**  $\frac{17}{55}$   
**F**  $\frac{23}{55}$

- 23** A function  $f(x)$  is defined for all real  $x$ , is differentiable at  $x = 0$ , and satisfies the equation:

$$f(x) = 2f\left(\frac{x}{3}\right) + x^2 + x$$

What is the value of  $f(7)$ ?

- A** 21
  - B** 56
  - C** 63
  - D** 84
  - E** 168
- 24** The function  $f(x)$  is defined by

$$f(x) = \frac{5^x}{5^x + \sqrt{5}}$$

What is the value of the following sum?

$$S = \sum_{r=0}^{100} f\left(\frac{r}{100}\right)$$

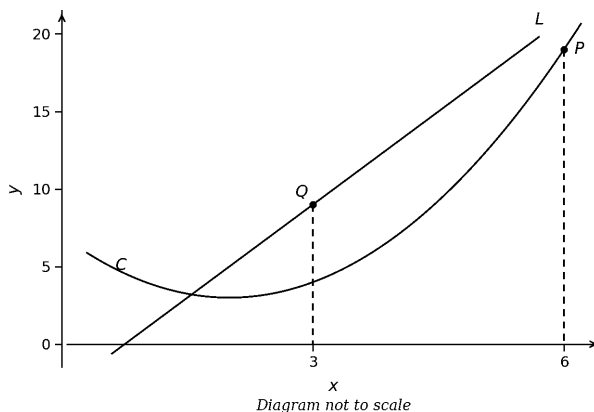
- A** 49.5
- B** 50
- C** 50.5
- D** 51
- E** 100
- F** 101

- 25** The diagram shows part of the curve  $C$  with equation  $y = (x - 2)^2 + 3$  and part of the straight line  $L$  with equation  $y = 4x + c$ , where  $c$  is a constant.

The point  $P$  lies on  $C$  at  $x = 6$ , and the point  $Q$  lies on  $L$  at  $x = 3$ .

The  $y$ -coordinate of  $P$  is equal to twice the  $y$ -coordinate of  $Q$  plus 1.

What is the  $y$ -coordinate of the point on  $L$  with  $x$ -coordinate 5?



- A**     $-3$   
**B**     $9$   
**C**     $17$   
**D**     $17.5$   
**E**     $23$   
**F**     $25$   
**G**     $47$
- 26** A right circular cone has base radius  $r$  and perpendicular height  $h$ . The volume of the cone is  $V$  and its total surface area is  $A$ . These quantities are related by the equation:

$$\ln(9V) - \ln(A) = \ln(r)$$

Which of the following expresses  $h$  in terms of  $r$ ?

- A**     $h = \frac{3}{4}r$   
**B**     $h = \frac{1}{2}r$   
**C**     $h = \frac{3}{8}r$   
**D**     $h = \frac{4}{3}r$   
**E**     $h = \frac{9}{40}r$   
**F**     $h = \frac{\sqrt{2}}{4}r$

**27** Let  $P$  and  $Q$  be positive real numbers.

$Q$  is the value obtained after  $P$  is first increased by 100%, then the result is decreased by 75%, and finally that result is increased by 100%.

What is the value of  $\log_4(Q) - \log_4(P)$ ?

- A**  $-1$
- B**  $-\frac{1}{2}$
- C**  $0$
- D**  $\frac{1}{2}$
- E**  $1$

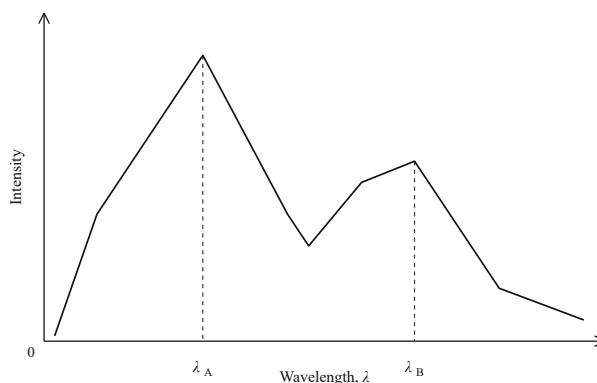
## Physics

- 1 The graph shows the current through a filament lamp at different voltages. The lamp is connected in series with a  $10\ \Omega$  resistor to a  $7.0\ \text{V}$  d.c. supply.

What is the power transferred to the lamp?

- A**  $0.90\ \text{W}$
- B**  $1.2\ \text{W}$
- C**  $1.6\ \text{W}$
- D**  $2.1\ \text{W}$
- E**  $2.8\ \text{W}$

- 2 The figure shows the intensity of an electromagnetic signal as a function of its wavelength,  $\lambda$ .



The signal has two distinct intensity peaks: a taller peak at wavelength  $\lambda_A$  and a shorter peak at a longer wavelength  $\lambda_B$ . This signal is then re-plotted to show its intensity as a function of frequency,  $f$ . Let  $f_A$  be the frequency corresponding to  $\lambda_A$  and  $f_B$  be the frequency corresponding to  $\lambda_B$ . Which statement correctly describes the new graph?

- A** The peak originally at  $\lambda_A$  is at a higher frequency than the peak from  $\lambda_B$ , and it is the taller peak.
- B** The peak originally at  $\lambda_A$  is at a lower frequency than the peak from  $\lambda_B$ , and it is the taller peak.
- C** The peak originally at  $\lambda_A$  is at a higher frequency than the peak from  $\lambda_B$ , but it is the shorter peak.
- D** The peak originally at  $\lambda_A$  is at a lower frequency than the peak from  $\lambda_B$ , and it is the shorter peak.
- E** The two peaks will appear at the same frequency on the new graph.
- F** The new frequencies are given by the relations  $f_A = c\lambda_A$  and  $f_B = c\lambda_B$ .

- 3 The graph shows the voltage from a simple a.c. generator whose coil rotates at constant speed. One voltage cycle corresponds to one complete rotation.

The rotation speed is doubled and the magnetic field strength is left unchanged. The coil starts in the same position and rotates in the same direction.

What is the first time after  $t = 0$  at which the new voltage reaches a positive maximum, and how does the new peak voltage compare with the old peak voltage?

- A 1.0 ms; smaller
- B 1.0 ms; larger
- C 2.0 ms; unchanged
- D 4.0 ms; larger
- E 8.0 ms; unchanged

- 4 A fixed mass of an ideal gas is kept at a constant temperature inside a cylinder fitted with a piston. The piston is pulled outwards so that the volume of the gas increases at a constant rate.

Which statement best describes the pressure of the gas?

- A It is constant.
- B It is increasing at a constant rate.
- C It is increasing at a decreasing rate.
- D It is decreasing at a constant rate.
- E It is decreasing at a decreasing rate.

- 5 Two solid spheres, P and Q, fall vertically through a fluid of density  $\rho$ . Sphere P has radius  $r$  and density  $3\rho$ ; sphere Q has radius  $2r$  and density  $2\rho$ .

For this question, the upthrust equals the weight of displaced fluid, and the volume of a sphere is proportional to the cube of its radius. The resistive force on a sphere of actual radius  $s$  is  $F = ksv$ , where  $v$  is its speed and  $k$  is constant.

Both spheres reach terminal velocities  $v_P$  and  $v_Q$ . What is  $v_Q/v_P$ ?

- A 1
- B 2
- C  $\frac{8}{3}$
- D 4
- E 8

- 6** An electric motor has an electrical power input of 160 W. The motor lifts a mass of 8.0 kg vertically at a constant speed of  $1.5 \text{ m s}^{-1}$ .

What is the efficiency of the motor? (Assume the acceleration due to gravity  $g = 10 \text{ m s}^{-2}$ .)

- A** 7.5%
- B** 15%
- C** 25%
- D** 40%
- E** 50%
- F** 75%

- 7** A station sends a microwave signal to a satellite 36000 km away. The satellite immediately sends an infrared signal to a second station 45000 km away from it.

Treat both paths as vacuum and ignore processing time. The speed of light is  $3.0 \times 10^8 \text{ m s}^{-1}$ .

What is the shortest time between emission at the first station and reception at the second?

- A** 0.030 s
- B** 0.12 s
- C** 0.15 s
- D** 0.27 s
- E** 0.54 s

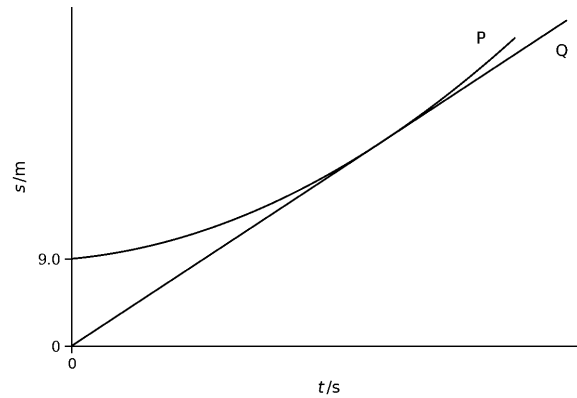
- 8 The displacement–time ( $s$ – $t$ ) graph shows the motion of two particles, P and Q, travelling in the same direction along a straight line.

Particle P starts at time  $t = 0$  from displacement  $s = 9.0$  m with an initial velocity of  $1.0 \text{ m s}^{-1}$  and moves with a constant acceleration of  $2.0 \text{ m s}^{-2}$ .

Particle Q starts at time  $t = 0$  from displacement  $s = 0$  and moves with a constant speed  $v$ .

On the graph, the straight line representing the motion of Q is tangent to the curve representing the motion of P.

What is the value of  $v$ ?



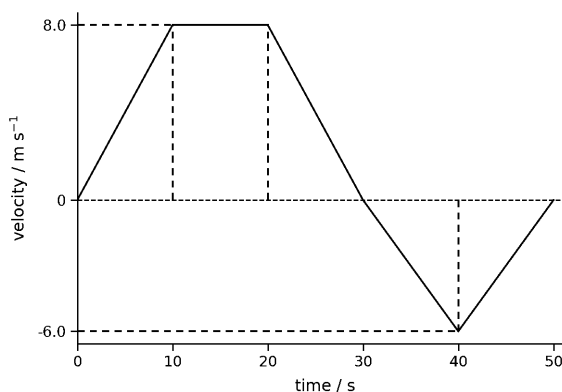
- A  $3.0 \text{ m s}^{-1}$
- B  $5.0 \text{ m s}^{-1}$
- C  $6.0 \text{ m s}^{-1}$
- D  $7.0 \text{ m s}^{-1}$
- E  $8.0 \text{ m s}^{-1}$
- F  $9.0 \text{ m s}^{-1}$
- G  $12.0 \text{ m s}^{-1}$
- H  $13.0 \text{ m s}^{-1}$

- 9** A small parcel of liquid at the bottom of a container is heated by a source below it. The parcel expands and begins to rise, initiating a convection current. Which statement best explains the physical origin of the net upward force acting on this heated parcel?
- A** The temperature of the parcel increases, giving the molecules sufficient kinetic energy to overcome gravity and move upwards.
  - B** The density of the parcel decreases, causing the weight of the surrounding fluid it displaces to exceed the weight of the parcel itself.
  - C** The volume of the parcel increases, causing the internal pressure of the parcel to become significantly greater than the external pressure from the surrounding fluid.
  - D** The density of the parcel increases, resulting in a stronger reaction force from the container base that pushes the parcel upwards.
  - E** The mass of the parcel decreases due to the expansion, reducing the gravitational force acting on it.

- 10** The graph shows how the velocity of a test vehicle travelling along a straight horizontal track varies with time over a period of 50 s.

What is the total distance travelled by the vehicle, its distance from its starting position at  $t = 50$  s, and the magnitude of its average velocity over the 50 s period?

|   | Total distance travelled / m | Distance from starting position / m | Magnitude of average velocity / $\text{m s}^{-1}$ |
|---|------------------------------|-------------------------------------|---------------------------------------------------|
| A | 100                          | 100                                 | 2.0                                               |
| B | 100                          | 100                                 | 4.4                                               |
| C | 100                          | 220                                 | 2.0                                               |
| D | 100                          | 220                                 | 4.4                                               |
| E | 220                          | 100                                 | 2.0                                               |
| F | 220                          | 100                                 | 4.4                                               |
| G | 220                          | 220                                 | 2.0                                               |
| H | 220                          | 220                                 | 4.4                                               |



- 11** An electrical heater with a power output of 100 W is used to melt a large block of ice at  $0^\circ\text{C}$ . The ice melts at a constant rate of  $0.20 \text{ g s}^{-1}$ .

What is the rate at which thermal energy is lost to the surroundings? (specific latent heat of fusion of ice =  $3.5 \times 10^5 \text{ J kg}^{-1}$ )

- A 30 W
- B 40 W
- C 70 W
- D 100 W
- E 170 W
- F 70 kW

- 12** An oscilloscope displays the signal from a microphone. For each sound, the trace measurements and oscilloscope settings are given in the table. The microphone's voltage amplitude is proportional to the sound-wave amplitude.

Compared with sound P, sound Q has:

| Sound | Divisions per cycle | Time / ms per division | Peak height / divisions | Voltage / V per division |
|-------|---------------------|------------------------|-------------------------|--------------------------|
| P     | 4.0                 | 0.50                   | 3.0                     | 0.20                     |
| Q     | 2.0                 | 0.50                   | 2.0                     | 0.30                     |

- A** half the frequency and the same amplitude
- B** the same frequency and half the amplitude
- C** the same frequency and the same amplitude
- D** double the frequency and double the amplitude
- E** double the frequency and the same amplitude
- 13** A sample initially contains 120 unstable nuclei of one isotope. Each nucleus decays directly to a stable nucleus. After one half-life, 72 unstable nuclei remain.
- Which statement is correct?
- A** The observation proves that the quoted half-life is wrong.
- B** Exactly 36 of the remaining nuclei must decay during the next half-life.
- C** The 72 surviving nuclei are less likely to decay because they have already survived one half-life.
- D** The expected number remaining after the next half-life is 36, but the actual number may differ.
- E** The next decay must occur after exactly one seventy-second of a half-life.

- 14** A ray of light is directed horizontally towards a plane mirror  $M_1$  which is inclined at an angle of  $30^\circ$  to the horizontal. The reflected ray strikes a second plane mirror  $M_2$  and, after reflecting from  $M_2$ , travels vertically upwards, as shown in the diagram.

Mirror  $M_1$  is now rotated clockwise through an angle  $\alpha$ , where  $\alpha < 30^\circ$ . The direction of the incident ray is unchanged.

In what direction and through what angle should mirror  $M_2$  be rotated in order for the ray to still emerge vertically upwards after reflecting from mirror  $M_2$ ?

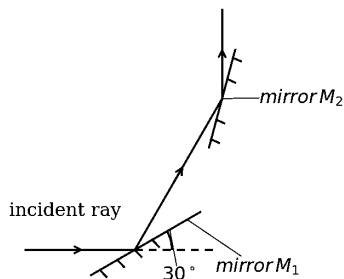


Diagram not to scale

- A clockwise through an angle  $\alpha$
  - B anticlockwise through an angle  $\alpha$
  - C clockwise through an angle  $2\alpha$
  - D anticlockwise through an angle  $2\alpha$
  - E clockwise through an angle  $\frac{\alpha}{2}$
  - F anticlockwise through an angle  $\frac{\alpha}{2}$
- 15** A ball is projected vertically upwards from the ground. It rises to a maximum height  $H$  and then falls back to the ground. Throughout the motion, the ball is subject to a force of air resistance that increases as the speed of the ball increases.

Let  $W_{\text{up}}$  be the magnitude of the work done by air resistance during the ascent.

Let  $W_{\text{down}}$  be the magnitude of the work done by air resistance during the descent.

Which of the following statements correctly compares  $W_{\text{up}}$  and  $W_{\text{down}}$ ?

- A  $W_{\text{up}} = W_{\text{down}}$ , because the distance travelled is the same in both phases.
- B  $W_{\text{up}} > W_{\text{down}}$ , because the speed at any given height is greater during the ascent.
- C  $W_{\text{up}} < W_{\text{down}}$ , because the time taken for the descent is greater than for the ascent.
- D  $W_{\text{up}} < W_{\text{down}}$ , because the ball accelerates due to gravity during the descent.

- 16** A periodic wave travels through a medium with a speed of  $5.0 \text{ m/s}$ . The period of the wave is  $2.0 \text{ s}$ .
- What is the number of complete wavelengths that fit into a distance of  $120 \text{ m}$  along the direction of propagation?
- A** 12
  - B** 24
  - C** 48
  - D** 60
- 17** A variable resistor is connected to a power supply that provides a constant voltage. The resistance of the resistor decreases at a constant rate over a period of time, but remains positive.
- Which statement correctly describes the power dissipated by the resistor during this time?
- A** It is increasing at an increasing rate.
  - B** It is increasing at a constant rate.
  - C** It is increasing at a decreasing rate.
  - D** It is not changing.
  - E** It is decreasing at a constant rate.
  - F** It is decreasing.
- 18** A sound wave travels to the right through a long tube of air. A tiny marked air particle is initially at P.
- Which description of the particle's motion is correct as the wave passes?
- A** It travels to the right with the wave and stops far from P.
  - B** It oscillates up and down, perpendicular to the tube.
  - C** It moves back and forth along the tube, with no net transport over a complete oscillation.
  - D** It remains stationary while the surrounding air carries the wave.
  - E** It follows a circular path whose radius is the wavelength.

- 19** Two identical light springs obey Hooke's law. In arrangement X they are in series; in arrangement Y they are in parallel. The same load of weight  $W$  is supported at rest in each arrangement.
- What is the ratio of the total elastic energy stored in X to that stored in Y?

- A** 1 : 4
- B** 1 : 2
- C** 1 : 1
- D** 2 : 1
- E** 4 : 1

- 20** An initially uncharged insulating object loses  $5.0 \times 10^{10}$  electrons when rubbed. It later gains  $2.0 \times 10^{10}$  electrons.

What is its final charge? The magnitude of the charge on one electron is  $1.6 \times 10^{-19}$  C.

- A**  $-11.2 \text{ nC}$
- B**  $-4.8 \text{ nC}$
- C**  $+3.2 \text{ nC}$
- D**  $+4.8 \text{ nC}$
- E**  $+11.2 \text{ nC}$

- 21** Two stationary coils share an iron core. The graph shows the current in coil P. A voltmeter is connected across coil Q.
- Increasing the current in P produces a positive voltage in Q. The voltmeter connections are fixed.
- During which labelled intervals is the voltage in Q non-zero?
- A** I and II only  
**B** all four intervals  
**C** I, III and IV only  
**D** II and IV only  
**E** I and III only
- 22** A resistor of constant resistance dissipates  $100\text{ W}$  when the potential difference across it is  $V$ . The potential difference is reduced to  $0.8V$ . What is the new power dissipated by the resistor?
- A**  $64\text{ ext W}$   
**B**  $80\text{ ext W}$   
**C**  $100\text{ ext W}$   
**D**  $125\text{ ext W}$
- 23** A progressive wave of frequency  $5\text{ Hz}$  travels from point P to point Q. The distance from P to Q is  $6\text{ m}$ . An observer at Q detects a wave peak  $0.10\text{ s}$  after an observer at P detects a peak. If the wavelength is known to be between  $3\text{ m}$  and  $5\text{ m}$ , what is the speed of the wave?
- A**  $12\text{ m/s}$   
**B**  $20\text{ m/s}$   
**C**  $30\text{ m/s}$   
**D**  $60\text{ m/s}$

- 24** Two spherical organisms, X and Y, have the same density and specific heat capacity. The radius of organism Y is three times the radius of organism X ( $r_Y = 3r_X$ ). Both are exposed to a radiant heat source that provides uniform energy per unit surface area.

Which option correctly identifies the ratio of total heat energy absorbed per unit time by Y compared to X, and the ratio of the initial rate of temperature rise of Y compared to X?

- A** Energy absorbed: 3 : 1; Rate of temperature rise: 1 : 3
  - B** Energy absorbed: 9 : 1; Rate of temperature rise: 3 : 1
  - C** Energy absorbed: 9 : 1; Rate of temperature rise: 1 : 3
  - D** Energy absorbed: 27 : 1; Rate of temperature rise: 1 : 1
  - E** Energy absorbed: 27 : 1; Rate of temperature rise: 1 : 9
- 25** A solid cuboid has a volume of  $0.0040 \text{ m}^3$  and a uniform density of  $2000 \text{ kg m}^{-3}$ . The areas of its three distinct faces are in the ratio 1 : 2 : 8.

What is the maximum pressure the cuboid can exert when it rests on a horizontal surface? (Use gravitational field strength  $g = 10 \text{ N kg}^{-1}$ )

- A** 100 Pa
  - B** 800 Pa
  - C** 1000 Pa
  - D** 4000 Pa
  - E** 8000 Pa
- 26** A sealed, fixed-volume vessel containing an ideal gas is placed in a large room of constant temperature and allowed to cool. The rate of heat loss from the gas is proportional to the temperature difference between the gas and the room. At time  $t = 0$ , the pressure of the gas is 240 kPa. At  $t = 20$  minutes, the pressure is 120 kPa. The half-life of the excess pressure is 10 minutes. The excess pressure is the difference between the gas pressure and its final equilibrium pressure.

What are the final equilibrium pressure of the gas and the pressure at  $t = 30$  minutes?

- A**  $P_f = 80 \text{ kPa}$ ;  $P(30) = 90 \text{ kPa}$
- B**  $P_f = 80 \text{ kPa}$ ;  $P(30) = 120 \text{ kPa}$
- C**  $P_f = 120 \text{ kPa}$ ;  $P(30) = 135 \text{ kPa}$
- D**  $P_f = 80 \text{ kPa}$ ;  $P(30) = 100 \text{ kPa}$
- E**  $P_f = 0 \text{ kPa}$ ;  $P(30) = 60 \text{ kPa}$

- 27** An ideal transformer has 800 turns on its primary coil and 40 on its secondary coil. The primary is supplied at 240 V a.c.

Identical lamps rated at 12 V, 6.0 W are connected in parallel across the secondary. The primary current must not exceed 0.15 A.

What is the largest number of lamps that can operate at their rated power?

- A** 3
- B** 4
- C** 5
- D** 6
- E** 12

## Chemistry

- 1 Four metals are labelled Q, R, S, and T. The following observations are made regarding their chemical properties and extraction methods:

1. Q and R are extracted by the electrolysis of their molten compounds, whereas S and T are extracted by heating their oxides with carbon.
2. S reacts with dilute hydrochloric acid to release a gas, but T does not react.
3. When metal Q is added to an aqueous solution of a salt of R, no reaction occurs.

Which of the following shows the metals arranged in order of decreasing reactivity (most reactive first)?

- A**  $Q > R > S > T$   
**B**  $R > Q > S > T$   
**C**  $R > Q > T > S$   
**D**  $S > T > R > Q$   
**E**  $T > S > Q > R$

- 2 Crude oil is separated into fractions using a fractionating column. The boiling point ranges for three of these fractions are given.

| Fraction | Boiling point range / °C |
|----------|--------------------------|
| Petrol   | 40 – 200                 |
| Kerosene | 180 – 250                |
| Diesel   | 250 – 350                |

Based on this information, which of the following statements is/are correct?

1. Diesel is collected at a higher temperature and lower down the fractionating column than petrol.
2. The intermolecular forces between molecules in the kerosene fraction are, on average, stronger than those in the petrol fraction.
3. The average number of carbon atoms per molecule is smaller in the diesel fraction than in the petrol fraction.

- A** none of them  
**B** 1 only  
**C** 2 only  
**D** 3 only  
**E** 1 and 2 only  
**F** 1 and 3 only  
**G** 2 and 3 only  
**H** 1, 2 and 3

- 3 Consider the following three gaseous ions:



Which of the following statements is/are correct?

- 1 All three ions have the electron configuration 2,8
- 2  $^{18}\text{O}^{2-}$  and  $^{20}\text{Ne}^+$  contain the same number of neutrons
- 3 The neutron-to-proton ratio in  $^{18}\text{O}^{2-}$  is greater than that in  $^{23}\text{Na}^+$

- A** none of them  
**B** 1 only  
**C** 2 only  
**D** 3 only  
**E** 1 and 2 only  
**F** 1 and 3 only  
**G** 2 and 3 only  
**H** 1, 2 and 3

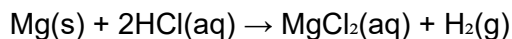
- 4 A reversible reaction producing a gaseous product was investigated in a closed vessel. The table below shows the percentage yield of the product at equilibrium and the time taken to reach equilibrium under different experimental conditions:

| Experiment | Temperature / K | Pressure / kPa | Catalyst | Equilibrium yield / % | Time to reach equilibrium / min |
|------------|-----------------|----------------|----------|-----------------------|---------------------------------|
| 1          | 400             | 100            | Absent   | 35                    | 45                              |
| 2          | 500             | 100            | Absent   | 18                    | 15                              |
| 3          | 400             | 200            | Absent   | 52                    | 20                              |
| 4          | 400             | 100            | Present  | 35                    | 12                              |

Which of the following equations could represent this reaction?

- A**  $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}) \quad \Delta H = -92 \text{ kJ mol}^{-1}$   
**B**  $2\text{SO}_3(\text{g}) \rightleftharpoons 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \quad \Delta H = +198 \text{ kJ mol}^{-1}$   
**C**  $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g}) \quad \Delta H = -9 \text{ kJ mol}^{-1}$   
**D**  $3\text{O}_2(\text{g}) \rightleftharpoons 2\text{O}_3(\text{g}) \quad \Delta H = +285 \text{ kJ mol}^{-1}$   
**E**  $2\text{O}_3(\text{g}) \rightleftharpoons 3\text{O}_2(\text{g}) \quad \Delta H = -285 \text{ kJ mol}^{-1}$   
**F**  $\text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{H}_2(\text{g}) \quad \Delta H = -41 \text{ kJ mol}^{-1}$

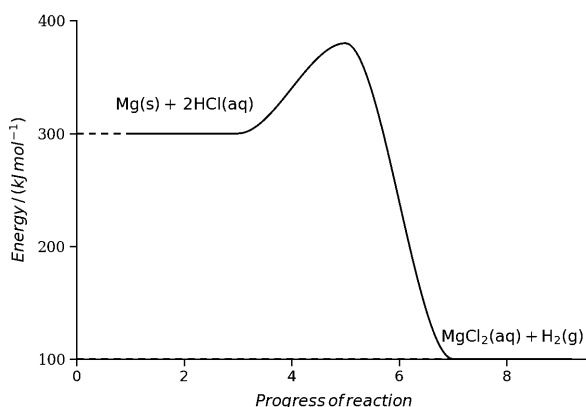
- 5 The reaction profile for the reaction between magnesium and excess dilute hydrochloric acid is shown below:



A sample of magnesium is added to an insulated container containing 150 g of dilute hydrochloric acid. Only 75% of the energy released by the reaction is absorbed by the solution, with the remainder transferred to the surroundings.

What mass of magnesium is required to increase the temperature of the solution from 18.0°C to 43.0°C?

(Assume that the specific heat capacity of the solution is  $4.0 \text{ J g}^{-1} \text{ } ^\circ\text{C}^{-1}$  and  $A_r(\text{Mg}) = 24.0$ .)



- A 1.35 g
- B 1.80 g
- C 2.40 g
- D 3.20 g
- E 4.80 g
- F 6.00 g

- 6 An element M forms an ionic oxide with the formula  $M_2O_3$ .

Which row in the table gives the correct chemical formulae for the sulfate, phosphate, and nitride of M?

|   | Sulfate of M  | Phosphate of M | Nitride of M |
|---|---------------|----------------|--------------|
| A | $MSO_4$       | $M_3(PO_4)_2$  | MN           |
| B | $MSO_4$       | $MPO_4$        | $M(NO_3)_3$  |
| C | $M(SO_4)_3$   | $MPO_4$        | MN           |
| D | $M(SO_4)_3$   | $M_3(PO_4)_3$  | $M_3N$       |
| E | $M_2(SO_4)_3$ | $MPO_4$        | MN           |
| F | $M_2(SO_4)_3$ | $MPO_4$        | $M_3N$       |
| G | $M_2(SO_4)_3$ | $M_3(PO_4)_2$  | MN           |
| H | $M_2(SO_4)_3$ | $M_3(PO_4)_2$  | $M(NO_3)_3$  |

- 7 Three aqueous solutions at  $25^\circ\text{C}$  are mixed together:

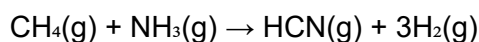
- $100\text{ cm}^3$  of  $0.050\text{ mol dm}^{-3}$   $\text{H}_2\text{SO}_4(\text{aq})$
- $150\text{ cm}^3$  of  $0.100\text{ mol dm}^{-3}$   $\text{HNO}_3(\text{aq})$
- $250\text{ cm}^3$  of  $0.040\text{ mol dm}^{-3}$   $\text{Ba}(\text{OH})_2(\text{aq})$

Assume that  $\text{H}_2\text{SO}_4$ ,  $\text{HNO}_3$ , and  $\text{Ba}(\text{OH})_2$  are completely dissociated in solution, and that volumes are additive.

What is the pH of the resulting mixture at  $25^\circ\text{C}$ ?

- A 1.0
- B 2.0
- C 3.0
- D 7.0
- E 11.0
- F 12.0
- G 13.0

- 8 The gas-phase reaction between methane and ammonia produces hydrogen cyanide and hydrogen according to the following equation:



The enthalpy change for this reaction is  $\Delta H = +255 \text{ kJ mol}^{-1}$ .

| Bond | Mean bond enthalpy / $\text{kJ mol}^{-1}$ |
|------|-------------------------------------------|
| C-H  | 415                                       |
| N-H  | 390                                       |
| H-H  | 435                                       |

Using the data provided, what is the calculated bond enthalpy of the  $\text{C} \equiv \text{N}$  triple bond in HCN?

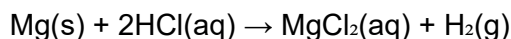
- A  $600 \text{ kJ mol}^{-1}$   
 B  $855 \text{ kJ mol}^{-1}$   
 C  $1110 \text{ kJ mol}^{-1}$   
 D  $1270 \text{ kJ mol}^{-1}$   
 E  $1365 \text{ kJ mol}^{-1}$   
 F  $1725 \text{ kJ mol}^{-1}$
- 9 An aqueous solution contains an equimolar mixture of  $\text{CuSO}_4$ ,  $\text{NaCl}$ , and  $\text{KNO}_3$ .

The solution is electrolysed using inert graphite electrodes.

Which row in the table correctly shows the initial half-equation occurring at the cathode (negative electrode) and at the anode (positive electrode)?

|   | Cathode half-equation                                                     | Anode half-equation                                                                                      |
|---|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| A | $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$ | $2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$                                |
| B | $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$ | $4\text{OH}^-(\text{aq}) \rightarrow \text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$ |
| C | $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$   | $2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$                                |
| D | $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$   | $4\text{OH}^-(\text{aq}) \rightarrow \text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$ |
| E | $\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow \text{Na}(\text{s})$     | $2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$                                |
| F | $\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow \text{Na}(\text{s})$     | $4\text{OH}^-(\text{aq}) \rightarrow \text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$ |
| G | $\text{K}^+(\text{aq}) + \text{e}^- \rightarrow \text{K}(\text{s})$       | $2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$                                |
| H | $\text{K}^+(\text{aq}) + \text{e}^- \rightarrow \text{K}(\text{s})$       | $4\text{OH}^-(\text{aq}) \rightarrow \text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$ |

- 10 Magnesium reacts with dilute hydrochloric acid according to the following equation:

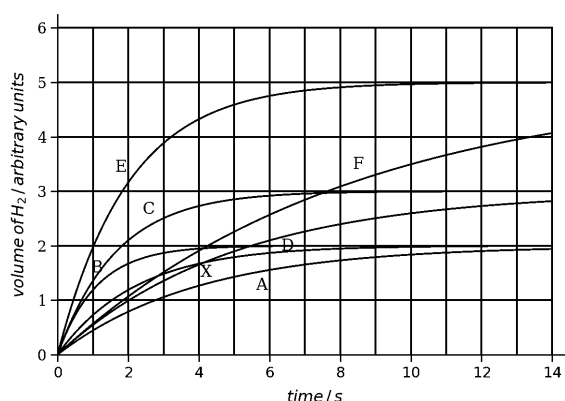


In Experiment 1, 0.48 g of magnesium ribbon was added to 100 cm<sup>3</sup> of 0.20 mol dm<sup>-3</sup> HCl(aq). Curve **X** on the graph shows how the volume of hydrogen gas produced changed with time at a constant temperature.

In Experiment 2, 0.36 g of identical magnesium ribbon was added to 50 cm<sup>3</sup> of 1.0 mol dm<sup>-3</sup> HCl(aq) at the same temperature and pressure.

Which curve (**A–F**) best represents how the volume of hydrogen gas produced changes with time in Experiment 2?

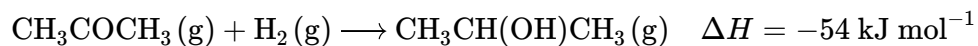
(Relative atomic mass,  $A_r$ : Mg = 24.0)



- A** A
- B** B
- C** C
- D** D
- E** E
- F** F

- 11 The structure of propan-2-ol is shown below.

Propan-2-ol can be synthesised by the gas-phase hydrogenation of propanone according to the following equation:



Mean bond energies:

- C—C = +347 kJ mol<sup>-1</sup>
- C—H = +413 kJ mol<sup>-1</sup>
- C—O = +358 kJ mol<sup>-1</sup>
- O—H = +464 kJ mol<sup>-1</sup>
- H—H = +436 kJ mol<sup>-1</sup>

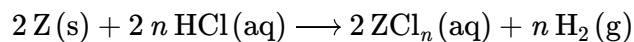
What is the mean bond energy of the C=O bond in propanone?



- A** 281 kJ mol<sup>-1</sup>
- B** 332 kJ mol<sup>-1</sup>
- C** 745 kJ mol<sup>-1</sup>
- D** 853 kJ mol<sup>-1</sup>
- E** 1181 kJ mol<sup>-1</sup>

- 12** The graph shows the first eight successive ionisation energies of a Period 3 element, Z.

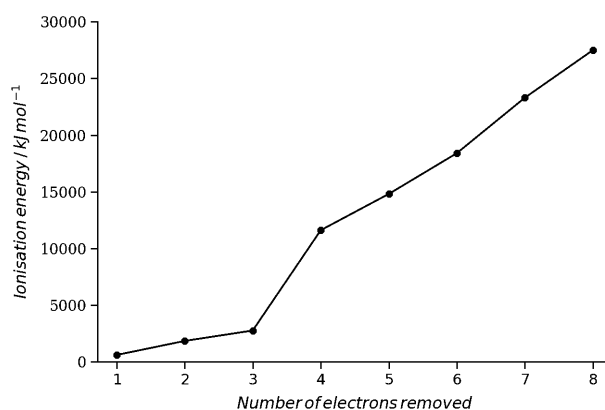
A 0.540 g sample of Z reacts completely with excess dilute hydrochloric acid according to the equation:



where  $n$  is the number of valence electrons in an atom of Z.

Given that the relative atomic mass of Z is 27.0, what volume of hydrogen gas is produced at room temperature and pressure (r.t.p.)?

(Assume that 1.00 mol of any gas occupies 24.0 dm<sup>3</sup> at r.t.p.)

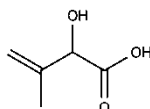


- A** 240 cm<sup>3</sup>
- B** 320 cm<sup>3</sup>
- C** 480 cm<sup>3</sup>
- D** 720 cm<sup>3</sup>
- E** 960 cm<sup>3</sup>
- F** 1440 cm<sup>3</sup>

**13** The skeletal structure of an organic compound, X, is shown below:

Which of the following statements about compound X is/are correct?

1. 1 mol of compound X reacts with an excess of sodium metal to produce 1 mol of  $\text{H}_2(\text{g})$ .
2. 1 mol of compound X reacts completely with 2 mol of aqueous sodium hydroxide (NaOH).
3. Compound X exhibits both optical isomerism and *E/Z* (geometric) isomerism.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 14** Two-dimensional thin-layer chromatography (2D-TLC) is used to separate a mixture of two amino acids, X and Y.

A sample of the mixture is spotted onto the origin near one corner of a square TLC plate.

1. The plate is placed in Solvent 1. The solvent front travels 12.0 cm vertically from the baseline. The plate is then removed and dried.
2. The plate is rotated by 90° and placed in Solvent 2. The new solvent front travels 10.0 cm perpendicular to the first direction.

The retention factors ( $R_f$ ) for the two amino acids in each solvent are shown in the table below:

| Amino acid | $R_f$ in Solvent 1 | $R_f$ in Solvent 2 |
|------------|--------------------|--------------------|
| X          | 0.65               | 0.20               |
| Y          | 0.40               | 0.60               |

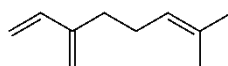
What is the straight-line distance between the centres of the spots for amino acid X and amino acid Y after the second run?

- A** 1.0 cm
- B** 3.0 cm
- C** 4.0 cm
- D** 5.0 cm
- E** 7.0 cm
- F** 10.0 cm
- G** 15.6 cm

- 15** The skeletal formula of the terpene myrcene is shown below:

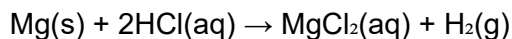
What is the minimum volume of hydrogen gas, measured at room temperature and pressure (rtp), required to completely hydrogenate  $0.34 \text{ cm}^3$  of liquid myrcene to form the fully saturated alkane 2,6-dimethyloctane?

( $M_r$  value: myrcene = 136. Density of myrcene =  $0.80 \text{ g cm}^{-3}$ . Molar volume of a gas at rtp =  $24.0 \text{ dm}^3 \text{ mol}^{-1}$ )



- A**  $0.048 \text{ cm}^3$
- B**  $0.096 \text{ cm}^3$
- C**  $0.144 \text{ cm}^3$
- D**  $0.288 \text{ cm}^3$
- E**  $0.048 \text{ dm}^3$
- F**  $0.096 \text{ dm}^3$
- G**  $0.144 \text{ dm}^3$
- H**  $0.288 \text{ dm}^3$

- 16** The reaction between magnesium and hydrochloric acid was investigated under different conditions at a constant temperature:

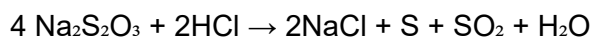
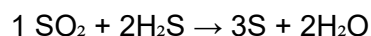


Which experiment (A–E) in the following table will produce 0.40 g of hydrogen gas in the shortest time?

( $A_r$  values: Mg = 24; H = 1.0)

|   | Magnesium form | Magnesium mass / g | HCl volume / cm <sub>3</sub> | HCl concentration / mol dm <sup>-3</sup> |
|---|----------------|--------------------|------------------------------|------------------------------------------|
| A | powder         | 2.4                | 250                          | 2.0                                      |
| B | powder         | 4.8                | 150                          | 2.0                                      |
| C | ribbon         | 4.8                | 100                          | 4.0                                      |
| D | powder         | 3.6                | 300                          | 1.5                                      |
| E | ribbon         | 4.8                | 200                          | 2.0                                      |

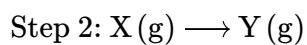
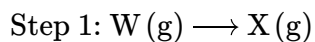
- 17** Four chemical equations involving sulfur compounds are shown below:



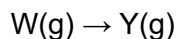
Which two equations represent disproportionation reactions?

- A** 1 and 2
- B** 1 and 3
- C** 1 and 4
- D** 2 and 3
- E** 2 and 4
- F** 3 and 4

18 A gas-phase reaction proceeds via a two-step mechanism:



The overall reaction is:



The following thermodynamic and kinetic values are given:

- Enthalpy change for Step 1:  $\Delta H_1 = +30 \text{ kJ mol}^{-1}$
- Activation energy for the forward reaction of Step 1:  $E_{a,1} = +80 \text{ kJ mol}^{-1}$
- Overall enthalpy change for  $W(g) \rightarrow Y(g)$ :  $\Delta H_{\text{overall}} = -40 \text{ kJ mol}^{-1}$
- Activation energy for the forward reaction of Step 2:  $E_{a,2} = +50 \text{ kJ mol}^{-1}$

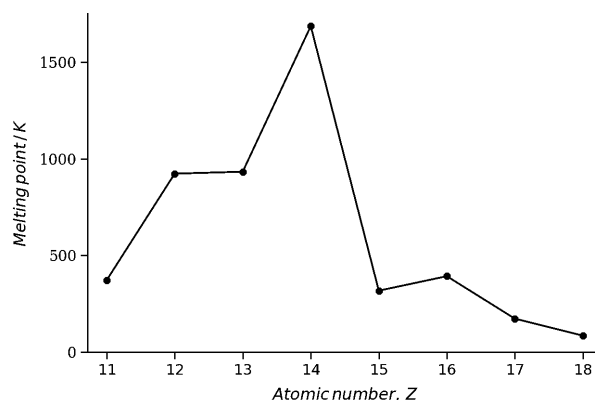
What is the enthalpy change for Step 2 ( $\Delta H_2$ ) and the activation energy for the reverse reaction of Step 2 ( $Y(g) \rightarrow X(g)$ )?

|   | Enthalpy change for Step 2 / $\text{kJ mol}^{-1}$ | Activation energy for reverse of Step 2 / $\text{kJ mol}^{-1}$ |
|---|---------------------------------------------------|----------------------------------------------------------------|
| A | -70                                               | +20                                                            |
| B | -70                                               | +80                                                            |
| C | -70                                               | +120                                                           |
| D | -10                                               | +40                                                            |
| E | -10                                               | +60                                                            |
| F | +70                                               | +20                                                            |
| G | +70                                               | +120                                                           |
| H | +10                                               | +60                                                            |

- 19** The graph shows the melting points of the eight consecutive elements across Period 3 of the Periodic Table (atomic numbers  $Z = 11$  to 18).

Which of the following statements correctly explain(s) features of this graph?

- 1 The sharp decrease in melting point from  $Z = 14$  to  $Z = 15$  occurs because the structure changes from a giant covalent network to a simple molecular structure.
- 2 The melting point of the element with  $Z = 16$  is higher than that of the element with  $Z = 15$  because  $S_8$  molecules have more electrons than  $P_4$  molecules, resulting in stronger London dispersion forces.
- 3 The element with  $Z = 18$  has the lowest melting point because it exists as diatomic molecules with fewer electrons per molecule than  $Cl_2$ .



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 20** Aqueous solutions of two different monoprotic acids, Acid X and Acid Y, are prepared. Both solutions have a concentration of  $0.10 \text{ mol dm}^{-3}$ .

The pH of the solution of Acid X is 1.0, and the pH of the solution of Acid Y is 3.0.

Which of the following statements is correct?

- A** Equal volumes of Acid X and Acid Y require the same volume of  $0.10 \text{ mol dm}^{-3}$  NaOH solution for complete neutralisation.
  - B** The solution of Acid Y has a higher electrical conductivity than the solution of Acid X.
  - C** If  $10 \text{ cm}^3$  of the solution of Acid Y is diluted to  $100 \text{ cm}^3$  with distilled water, the pH of the resulting solution will be 4.0.
  - D** The concentration of the anion  $\text{X}^-$  is approximately equal to the concentration of the anion  $\text{Y}^-$ .
  - E** When reacted with an excess of magnesium ribbon, the initial rate of hydrogen gas production is greater for Acid Y than for Acid X.
- 21** A neutral atom of element X has electrons in three occupied electron shells, and its  $+2$  ion has the electron configuration of neon.

Which row of the table correctly compares the elements immediately adjacent to X in the Periodic Table with element X?

|          | element immediately above X                 | element immediately below X                                    | element immediately to the left of X                              | element immediately to the right of X       |
|----------|---------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------|
| <b>A</b> | has a higher first ionisation energy than X | reacts more vigorously with cold water than X                  | its stable cation has a smaller ionic radius than $\text{X}^{2+}$ | has a higher first ionisation energy than X |
| <b>B</b> | has a lower first ionisation energy than X  | reacts more vigorously with cold water than X                  | its stable cation has a larger ionic radius than $\text{X}^{2+}$  | has a higher first ionisation energy than X |
| <b>C</b> | has a higher first ionisation energy than X | reacts more vigorously with cold water than X                  | its stable cation has a larger ionic radius than $\text{X}^{2+}$  | has a lower first ionisation energy than X  |
| <b>D</b> | has a higher first ionisation energy than X | forms a hydroxide that is less soluble in water than that of X | its stable cation has a larger ionic radius than $\text{X}^{2+}$  | has a higher first ionisation energy than X |
| <b>E</b> | has a lower first ionisation energy than X  | forms a hydroxide that is less soluble in water than that of X | its stable cation has a smaller ionic radius than $\text{X}^{2+}$ | has a lower first ionisation energy than X  |

- 22 The first six successive ionisation energies of a Period 3 element, Z, are shown below:

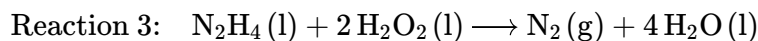
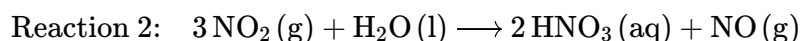
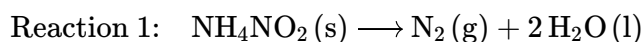
| Ionisation energy                       | 1st | 2nd  | 3rd  | 4th  | 5th   | 6th   |
|-----------------------------------------|-----|------|------|------|-------|-------|
| Value / $\text{kJ}\cdot\text{mol}^{-1}$ | 789 | 1577 | 3232 | 4356 | 16091 | 19785 |

Element Z reacts completely with excess chlorine gas to form a single liquid chloride.

Which row in the table correctly identifies the formula of this chloride, the molecular shape of this chloride, and the pH of the resulting mixture when this chloride is added to an excess of pure water?

|   | Formula of chloride | Molecular shape      | pH of resulting mixture |
|---|---------------------|----------------------|-------------------------|
| A | $\text{ZCl}_3$      | trigonal planar      | $<7$                    |
| B | $\text{ZCl}_3$      | pyramidal            | $>7$                    |
| C | $\text{ZCl}_4$      | square planar        | $<7$                    |
| D | $\text{ZCl}_4$      | square planar        | $=7$                    |
| E | $\text{ZCl}_4$      | tetrahedral          | $<7$                    |
| F | $\text{ZCl}_4$      | tetrahedral          | $=7$                    |
| G | $\text{ZCl}_5$      | trigonal bipyramidal | $<7$                    |
| H | $\text{ZCl}_5$      | tetrahedral          | $=7$                    |

- 23** Consider the following three balanced chemical equations involving nitrogen-containing compounds:



Which of the following statements is/are correct?

1. In reaction 1, nitrogen is both oxidised and reduced.
2. In reaction 2, nitrogen undergoes disproportionation.
3. In reaction 3, hydrogen peroxide acts as a reducing agent.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 24** Consider only the three Group 2 elements Mg, Ca, and Sr, and only the three Group 16 elements O, S, and Se.

Which of the following statements is/are correct for the compound strontium oxide?

- 1 It is formed from the Group 2 element with the lowest first ionisation energy.
- 2 It is formed from the Group 16 element with the highest electronegativity and the Group 2 element with the lowest reactivity with cold water.
- 3 It is formed from the Group 16 element with the lowest boiling point.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

**25** Gas-liquid chromatography was used to analyse an unknown liquid mixture.

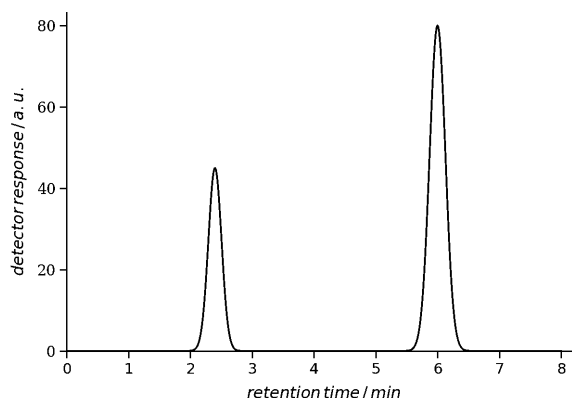
Samples of three pure reference compounds, X, Y, and Z, were each analysed separately under identical conditions (same stationary phase, column temperature, and carrier gas flow rate). Their retention times were recorded as follows:

- Compound X: 2.4 min
- Compound Y: 4.8 min
- Compound Z: 6.0 min

A sample of the unknown mixture was then injected under the exact same conditions, producing the chromatogram shown below.

Which of the following statements is/are correct?

- 1 The inert carrier gas passing through the column acts as the mobile phase.
- 2 The unknown mixture contains compounds X and Z, but does not contain compound Y.
- 3 Compound Z has a longer retention time than compound X because compound Z has a stronger affinity for the mobile phase than compound X.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 26** An oxyanion has the formula  $[\text{XO}_4]^{2-}$ , where X represents an isotope of an element with atomic number  $z$  and mass number  $2z + 6$ .

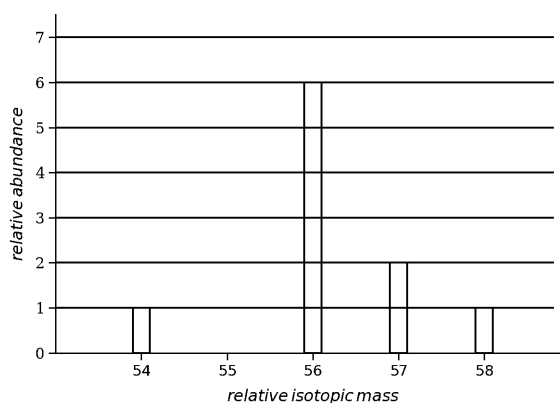
Each oxygen atom present in the ion is the isotope  $^{16}_8\text{O}$ .

Which row in the table gives the number of protons, neutrons, and electrons in this  $[\text{XO}_4]^{2-}$  ion?

|   | protons | neutrons | electrons |
|---|---------|----------|-----------|
| A | $z+32$  | $z+38$   | $z+30$    |
| B | $z+32$  | $z+38$   | $z+32$    |
| C | $z+32$  | $z+38$   | $z+34$    |
| D | $z+32$  | $2z+38$  | $z+34$    |
| E | $z+32$  | $2z+70$  | $z+34$    |
| F | $z+34$  | $z+38$   | $z+34$    |

- 27** The mass spectrum shows the relative abundances of the isotopes of an element.

What is the relative atomic mass ( $A_r$ ) of this element?



- A** 56.0  
**B** 56.2  
**C** 56.25  
**D** 56.3  
**E** 56.5

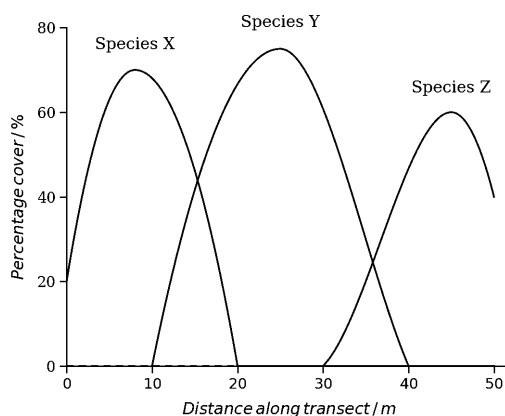
## **Biology**

- 1 An ecological investigation was carried out to study the distribution and abundance of three plant species (X, Y, and Z) along a coastal sand dune.

A continuous 50 m belt transect was established from the edge of the beach inland, using contiguous 1 m × 1 m quadrats (50 quadrats in total). The percentage cover of each plant species was recorded in each quadrat. The results are shown in the graph below.

Which of the following statements is/are correct?

- 1 The frequency of occurrence of species Y across the entire 50 m transect is 60%.
- 2 At the 8 m position along the transect, the population density of species X is definitely higher than at the 18 m position.
- 3 At the 35 m position along the transect, species Y and species Z together cover the entire ground area within the quadrat.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 2** Which of the following statements about the human circulatory system is/are correct?
1. The walls of capillaries consist of a single layer of cells, providing a short diffusion distance for substances.
  2. The pulmonary artery transports oxygenated blood from the heart to the lungs.
  3. During a cardiac cycle, the blood pressure in the aorta is significantly higher than the pressure in the pulmonary artery.
- A** none of them  
**B** 1 only  
**C** 2 only  
**D** 3 only  
**E** 1 and 2 only  
**F** 1 and 3 only  
**G** 2 and 3 only  
**H** 1, 2 and 3

- 3 An investigation was carried out to study the effect of salinity on the germination of a crop species.

Three identical Petri dishes were prepared, each containing 20 viable seeds placed on filter paper moistened with a different concentration of sodium chloride solution: 0 mM NaCl (control), 50 mM NaCl, and 100 mM NaCl.

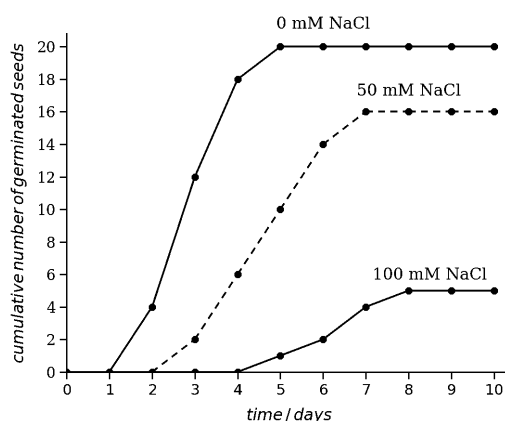
All other environmental variables, including temperature and light conditions, were kept constant.

The cumulative number of germinated seeds in each dish was recorded once every day for 10 days.

The results are shown in the graph.

Which of the following statements is/are correct?

- 1 Over the first 5 days, the average rate of germination was highest in the 0 mM NaCl solution.
- 2 At day 4, 30% of the seeds in the 50 mM NaCl solution remained ungerminated.
- 3 The reciprocal of the time taken for 50% of the seeds in a treatment to germinate can be used as a measure of the relative rate of germination.

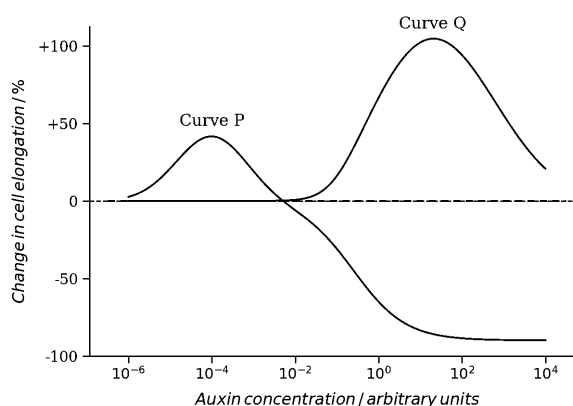


- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 4 The graph shows the effect of different concentrations of auxin (IAA) on cell elongation in two different plant tissues, P and Q, measured as a percentage change relative to untreated controls (0%).

Which of the following statements is/are correct?

- 1 Curve P represents the response of root tissue, and Curve Q represents the response of shoot tissue.
- 2 In a horizontal seedling where auxin accumulates on the lower side to 1.0 arbitrary units while the upper side remains at  $10^{-4}$  arbitrary units, differential elongation will cause the shoot to bend downwards.
- 3 In a vertical root with a uniform auxin concentration of 1.0 arbitrary units, applying an inhibitor that reduces the auxin concentration throughout the root to  $10^{-4}$  arbitrary units will increase the root elongation rate.



- A 1 only
- B 2 only
- C 3 only
- D 1 and 2 only
- E 1 and 3 only
- F 2 and 3 only
- G 1, 2 and 3
- H none of the statements

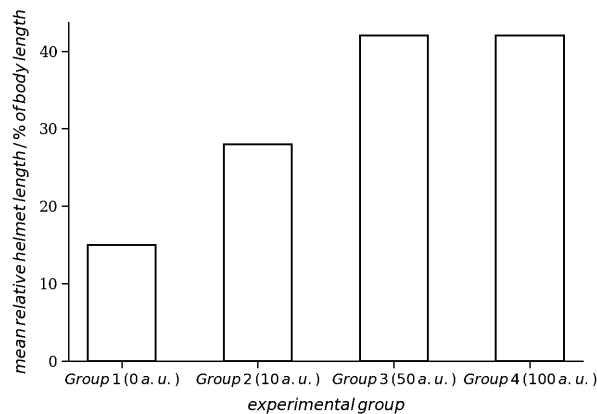
- 5 Genetically identical individuals of the water flea *Daphnia magna* were divided into four equal groups of twenty individuals. All four groups were raised in identical volumes of water, at the same temperature, with the same light cycle, and fed equal quantities of algae. The only difference between the groups was the concentration of a chemical cue (kairomone) released by a predator added to the water:

- Group 1: 0 a.u. of predator cue
- Group 2: 10 a.u. of predator cue
- Group 3: 50 a.u. of predator cue
- Group 4: 100 a.u. of predator cue

Upon reaching maturity, the mean relative helmet length (the length of the defensive helmet as a percentage of body length) was measured for each group. The results are shown in the graph.

Which of the following statements could explain these results?

- 1 The difference in mean relative helmet length between Group 1 and Group 2 is caused by genetic differences between the individuals in the two groups.
- 2 The identical mean relative helmet length in Group 3 and Group 4 could be due to an environmental factor, such as food availability, becoming limiting.
- 3 The identical mean relative helmet length in Group 3 and Group 4 could be due to reaching the maximum phenotypic limit determined by the genotype of the clone.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 6 A short section of mRNA has the following sequence:

5' - AUG CUC CAC UAA - 3'

The genetic code for selected amino acids is provided in the table below:

| Amino Acid      | Codons             |
|-----------------|--------------------|
| Histidine (His) | CAU, CAC           |
| Glutamine (Gln) | CAA, CAG           |
| Leucine (Leu)   | CUU, CUC, CUA, CUG |
| Valine (Val)    | GUU, GUC, GUA, GUG |

A mutation results in the substitution of the cytosine (C) at nucleotide position 6 with a guanine (G).

Which of the following describes the effect on the polypeptide sequence?

- A** The amino acid sequence remains unchanged.
- B** Leu is replaced by Val.
- C** His is replaced by Gln.
- D** Leu is replaced by His.
- 7 Epidermolysis bullosa is an inherited condition in humans caused by a mutation in a gene that codes for a structural protein needed to anchor the outer layer of skin (epidermis) to underlying tissues.

In an experimental gene therapy treatment, skin stem cells were taken from an adult patient with this condition. A functional copy of the gene was inserted into the nuclear DNA of these stem cells using a viral vector. The genetically modified stem cells were then grown in the laboratory to form sheets of epidermal tissue, which were grafted onto the patient's skin to heal open wounds.

Which of the following statements about this gene therapy treatment is/are correct?

- 1 Transcription and translation of the inserted gene in the cells of the grafted epidermis can produce the functional structural protein.
- 2 When the genetically modified stem cells divide by mitosis, the inserted gene is replicated so that daughter cells retain the inserted gene.
- 3 The patient's biological children will inherit the inserted functional gene from this patient.

- A** 1 only
- B** 2 only
- C** 3 only
- D** 1 and 2 only
- E** 1 and 3 only
- F** 2 and 3 only
- G** 1, 2 and 3

- 8 Two cellular structures, structure X and structure Y, were imaged using electron microscopy. The micrographs are not shown.

Some of the data collected is shown in the table.

|                                 | structure X | structure Y |
|---------------------------------|-------------|-------------|
| actual length / nm              | 2500        | 25          |
| image length on micrograph / mm | 50          | 1.5         |

Which of the following statements is/are correct?

- 1 Structure X has been magnified 20 000 times.
- 2 Structure X could be a mitochondrion and structure Y could be a ribosome.
- 3 Structure Y can be resolved using a standard light microscope.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 9 A sealed flask containing a suspension of yeast (*Saccharomyces cerevisiae*) in a glucose nutrient solution was incubated at 25 °C. The rate of oxygen uptake, the rate of carbon dioxide release, and the rate of ethanol production were measured after 1 hour and after 6 hours.

|                                                   | Time = 1 hour | Time = 6 hours |
|---------------------------------------------------|---------------|----------------|
| rate of O <sub>2</sub> uptake / arbitrary units   | 8.4           | 0.2            |
| rate of CO <sub>2</sub> release / arbitrary units | 8.5           | 5.6            |
| rate of ethanol production / arbitrary units      | 0.1           | 5.5            |

Which of the following statements is/are correct?

- 1 A single molecule of glucose metabolised by the yeast yielded fewer molecules of ATP at 6 hours than at 1 hour.
- 2 At 6 hours, the oxidation of NADH to NAD<sup>+</sup> was coupled to the reduction of oxygen at the inner mitochondrial membrane.
- 3 At 6 hours, some of the energy originally stored in glucose remained in an organic waste product released into the medium.

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 10** In a population of rats, resistance to the anticoagulant poison warfarin is determined by a single gene with two alleles,  $R$  and  $S$ .

The table provides information about the phenotypes associated with each genotype:

| Genotype | Effect of standard dose of warfarin | Health in the absence of warfarin         |
|----------|-------------------------------------|-------------------------------------------|
| $SS$     | fatal internal bleeding             | normal health and survival                |
| $RS$     | survives                            | normal health and survival                |
| $RR$     | survives                            | severe vitamin K deficiency (often fatal) |

Which of the following statements is/are correct?

- 1 In habitats where warfarin is never applied, natural selection is likely to maintain a low frequency of the  $R$  allele in the population.
- 2 The application of warfarin causes the  $S$  allele to mutate into the  $R$  allele in surviving rats.
- 3 If all  $SS$  individuals in a population are killed by warfarin before they reproduce, the  $S$  allele will be completely eliminated from the next generation.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 11 An agricultural investigation was carried out using two varieties of a crop plant, Variety X and Variety Y. For each variety, all plants used were genetically identical clones.

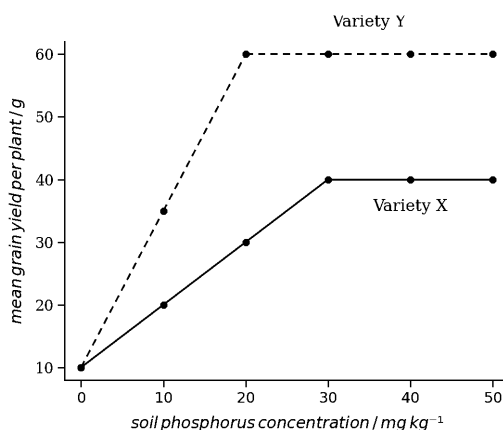
Groups of plants of each variety were grown across a range of soil phosphorus concentrations under strictly controlled environmental conditions:

- equal volumes of soil containing identical baseline nutrients
- identical temperature, light intensity, and photoperiod
- watered with equal volumes of water
- identical atmospheric CO<sub>2</sub> concentration

After 16 weeks of growth, the mean grain yield per plant was determined for each group. The results are shown in the graph below.

Which of the following statements is/are correct?

- 1 At a soil phosphorus concentration of 10 mg kg<sup>-1</sup>, the difference in mean grain yield between Variety X and Variety Y is due to differences in their genotype.
- 2 For Variety X, the difference in mean grain yield between plants grown at 10 mg kg<sup>-1</sup> and plants grown at 25 mg kg<sup>-1</sup> is due to differences in their genotype.
- 3 Above 30 mg kg<sup>-1</sup>, the plateau in mean grain yield for Variety X could be because another abiotic factor has become limiting.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 12** A botanist crosses a homozygous purple-flowered plant (genotype  $BB$ ) with a homozygous white-flowered plant (genotype  $bb$ ). Flower colour is controlled by a single autosomal gene, where the purple allele ( $B$ ) is dominant to the white allele ( $b$ ).

One of the offspring produced from this cross has entirely white flowers. Genetic analysis confirms that both parent plants are the true biological parents, and that all somatic cells and germ cells of this white-flowered offspring have the same genotype.

Which of the following statements could explain this result?

- 1 A loss-of-function mutation in the  $B$  allele occurred during meiosis in the  $BB$  parent.
- 2 A loss-of-function mutation in the  $B$  allele occurred during the first mitotic division of the embryo.
- 3 Chromosome non-disjunction during meiosis in the  $BB$  parent produced a gamete lacking the chromosome carrying the  $B$  gene.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 13** The table shows some mRNA codons and the corresponding amino acids or translation stop signals:

| Codon | Amino acid / function |
|-------|-----------------------|
| AUG   | Met                   |
| CCU   | Pro                   |
| CCG   | Pro                   |
| CGA   | Arg                   |
| CGU   | Arg                   |
| GAA   | Glu                   |
| GAG   | Glu                   |
| GAU   | Asp                   |
| UAA   | Stop                  |
| UAG   | Stop                  |
| UGC   | Cys                   |
| UGG   | Trp                   |
| UUA   | Leu                   |
| UUG   | Leu                   |
| UUC   | Phe                   |
| UUU   | Phe                   |

A synthetic mRNA molecule has the following sequence of 30 nucleotides, beginning with a start codon at the 5' end:

AUG UUC CGU GAA UGG CCU GAU UGC UUA GAG

Single-base substitution mutations took place at both the 8<sup>th</sup> and 14<sup>th</sup> nucleotide positions in this sequence.

Using **only** the information provided and translating the mRNA sequence from left to right, which of the following statements **could** be correct for the resulting polypeptide?

- 1 This polypeptide could be only four amino acids long.
- 2 The third amino acid in the polypeptide could be unaffected by the mutation.
- 3 If translation does not stop prematurely, the resulting polypeptide could contain seven different amino acids.

- A** none of them  
**B** 1 only  
**C** 2 only

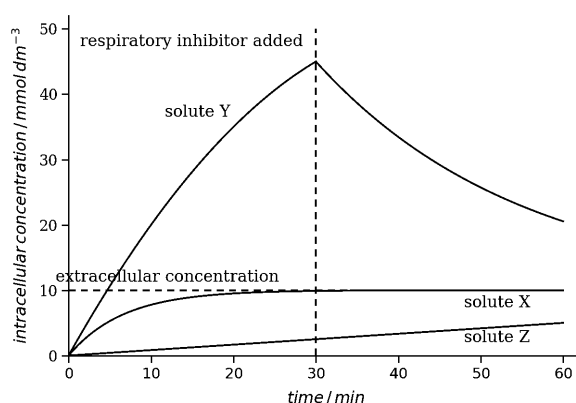
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 14** Cells of a unicellular eukaryote were placed into a solution containing three different uncharged solutes: X, Y, and Z. The concentration of each solute in the external solution was maintained at a constant  $10 \text{ mmol dm}^{-3}$  throughout the experiment.

The intracellular concentration of each solute was measured over 60 minutes. At 30 minutes, a respiratory inhibitor was added to the solution. The results are shown in the graph.

Which of the following statements is/are correct?

- 1 The accumulation of solute Y inside the cell between 0 and 30 minutes requires metabolic energy (ATP).
- 2 Solute X is transported into the cell against its concentration gradient.
- 3 After the addition of the respiratory inhibitor at 30 minutes, solute Y leaves the cell by moving down its concentration gradient.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 15** The allele frequencies in the gene pool of a sexually reproducing population can change over generations as a result of various biological processes and environmental events.

Which of the following would be expected to cause a change in allele frequencies in the gene pool of a population?

- 1 widespread application of a chemical herbicide across an agricultural weed population
- 2 immigration of fertile individuals from a genetically distinct neighbouring population
- 3 a severe population bottleneck caused by a natural disaster that randomly kills the majority of individuals
- 4 random mating occurring within a very large population in the absence of mutation, migration, and differential selection

- A** 1 and 2 only
- B** 1 and 4 only
- C** 2 and 3 only
- D** 3 and 4 only
- E** 1, 2 and 3 only
- F** 1, 2 and 4 only
- G** 2, 3 and 4 only
- H** 1, 2, 3 and 4

- 16** An experiment was conducted to investigate the breakdown of hydrogen peroxide catalysed by catalase extracted from human liver cells.

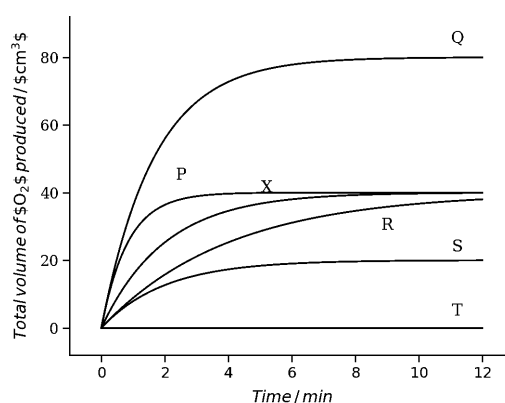
In a control experiment, 10 cm<sup>3</sup> of 1.0 mol dm<sup>-3</sup> H<sub>2</sub>O<sub>2</sub> was mixed with 1.0 cm<sup>3</sup> of human catalase solution at 25 °C and pH 7.0. The reaction was allowed to proceed to completion, and the total volume of oxygen gas produced was recorded over time, producing curve **X** on the graph.

The experiment was repeated under three separate modifications:

- **Modification 1:** Carried out at 37 °C instead of 25 °C.
- **Modification 2:** Carried out using 10 cm<sup>3</sup> of 0.5 mol dm<sup>-3</sup> H<sub>2</sub>O<sub>2</sub> at 25 °C.
- **Modification 3:** Carried out at 25 °C after boiling the catalase solution for 10 minutes and then cooling it.

Which row in the table correctly identifies the curve obtained for each modification?

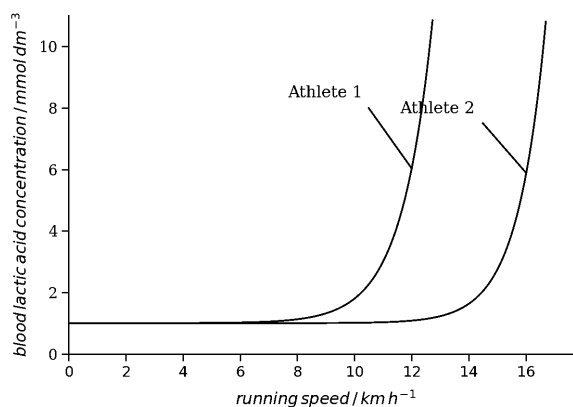
|          | Modification 1 | Modification 2 | Modification 3 |
|----------|----------------|----------------|----------------|
| <b>A</b> | P              | S              | T              |
| <b>B</b> | P              | R              | T              |
| <b>C</b> | P              | S              | R              |
| <b>D</b> | Q              | S              | T              |
| <b>E</b> | Q              | R              | T              |
| <b>F</b> | Q              | S              | R              |
| <b>G</b> | R              | S              | T              |
| <b>H</b> | R              | P              | T              |



- 17 The graph shows the blood lactic acid concentration of two athletes, Athlete 1 and Athlete 2, during an exercise test where running speed was increased incrementally in stages.

Which of the following statements correctly describe(s) the athletes?

- 1 At a running speed of  $11 \text{ km h}^{-1}$ , Athlete 1 relies more on anaerobic respiration to meet their energy demands than Athlete 2.
- 2 At a running speed of  $5 \text{ km h}^{-1}$ , the rate of lactic acid production in both athletes is zero.
- 3 If both athletes run at  $11 \text{ km h}^{-1}$  for 15 minutes, Athlete 1 will incur a larger oxygen debt than Athlete 2.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 18** Researchers investigated the role of Protein Q in the synthesis of a stress-response metabolite, M. Three bacterial strains were compared: a wild type, a mutant lacking the gene for Protein Q, and a strain engineered to contain high cellular levels of Protein Q. The concentration of metabolite M (arbitrary units) was measured in each strain after incubation at either 20°C or 40°C.

The results are shown in the table below:

| Strain     | Level of Protein Q | Concentration of M at 20°C | Concentration of M at 40°C |
|------------|--------------------|----------------------------|----------------------------|
| Wild Type  | Normal             | 5   100                    | 5   25                     |
| Mutant     | None               | 5   25                     | 5   25                     |
| Engineered | High               | 5   250                    | 5   250                    |

Which of the following conclusions is best supported by the data?

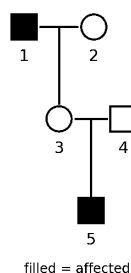
- A** Protein Q is essential for the initiation of metabolite M synthesis.
  - B** Protein Q acts as a temperature sensor that triggers the production of metabolite M.
  - C** The production of metabolite M is proportional to the concentration of Protein Q at both temperatures.
  - D** Protein Q enhances the production of metabolite M but only in the presence of the thermal stimulus.
  - E** The presence of Protein Q suppresses the basal production of metabolite M at 20°C.
  - F** Metabolite M synthesis is regulated solely by the concentration of Protein Q.
- 19** In a species of plant, the allele for tall stems ( *T* ) is dominant to the allele for dwarf stems ( *t* ). A tall plant is crossed with a dwarf plant. The resulting offspring show a 1:1 ratio of tall plants to dwarf plants.
- Two tall plants, each with the same genotype as the original tall parent, are then crossed with each other.
- What is the expected proportion of offspring from this second cross that will be dwarf?
- A** 0
  - B** 1/4
  - C** 1/2
  - D** 3/4

- 20** In mice, the allele for black fur,  $B$ , is dominant to the allele for brown fur,  $b$ .  
Two heterozygous black mice are crossed. A black offspring from this cross is selected at random and crossed with a heterozygous black mouse.  
What is the probability that their first offspring has brown fur?
- A** 0
  - B**  $\frac{1}{12}$
  - C**  $\frac{1}{6}$
  - D**  $\frac{1}{4}$
  - E**  $\frac{1}{3}$
  - F**  $\frac{1}{2}$
- 21** A diploid cell contains 12 chromosomes. Which row correctly gives the number of chromosomes in each daughter cell after mitosis and in each gamete after meiosis?
- A** Mitosis 6; meiosis 6
  - B** Mitosis 6; meiosis 12
  - C** Mitosis 12; meiosis 6
  - D** Mitosis 12; meiosis 12
  - E** Mitosis 24; meiosis 6
  - F** Mitosis 24; meiosis 12

- 22** The pedigree shows the inheritance of an X-linked recessive condition in a family.

Individual 5 has been diagnosed with Klinefelter syndrome (karyotype 47, XXY) and expresses the recessive condition.

Assuming no new mutations occurred and no crossing over took place between the gene locus and the centromere, in which parent did the non-disjunction event occur, and at which stage of cell division?



*Diagram not to scale*

- A** mother; meiosis I
  - B** mother; meiosis II
  - C** mother; mitosis after fertilisation
  - D** father; meiosis I
  - E** father; meiosis II
  - F** father; mitosis after fertilisation
  - G** either parent; meiosis I
  - H** either parent; meiosis II
- 23** A culture of secretory cells has a total tissue mass of 10 mg. The average mass of a single cell in the culture is 1 ng.
- Each cell synthesizes a specific protein at a rate of  $10^5$  molecules per second. The mass of one molecule of this protein is  $10^{-18}$  g.
- What is the total mass of the protein synthesized by the culture in 100 minutes?
- A** 0.1 mg
  - B** 1.0 mg
  - C** 6.0 mg
  - D** 10 mg
  - E** 60 mg
  - F** 6.0 g

- 24** A suspension contains a mixture of two strains of yeast: a wild-type strain that secretes the enzyme invertase, and a mutant strain that does not. The total concentration of yeast cells in the suspension is  $8.0 \times 10^6$  cells  $\text{mL}^{-1}$  and the invertase activity is 300 arbitrary units (AU)  $\text{mL}^{-1}$ .  
A separate control sample containing only the wild-type strain at a concentration of  $1.0 \times 10^6$  cells  $\text{mL}^{-1}$  has an invertase activity of 50 AU  $\text{mL}^{-1}$ .  
Assuming the mutant strain has zero invertase activity, what is the concentration of the mutant strain in the mixed suspension?
- A**  $1.0 \times 10^6$  cells  $\text{mL}^{-1}$
  - B**  $2.0 \times 10^6$  cells  $\text{mL}^{-1}$
  - C**  $4.0 \times 10^6$  cells  $\text{mL}^{-1}$
  - D**  $5.0 \times 10^6$  cells  $\text{mL}^{-1}$
  - E**  $6.0 \times 10^6$  cells  $\text{mL}^{-1}$

- 25** The diagram shows a circular plasmid vector containing an origin of replication (*ori*), an ampicillin resistance gene (*Amp<sup>R</sup>*), and a  $\beta$ -galactosidase gene (*lacZ*).

The enzyme  $\beta$ -galactosidase hydrolyses the colourless substrate X-gal into a blue product in the presence of the inducer IPTG.

Two recombinant plasmids were produced by inserting a target gene into one of the unique restriction sites:

- **Plasmid A** has the target gene inserted into the *EcoRI* site located within the coding sequence of *lacZ*.
- **Plasmid B** has the target gene inserted into the *PstI* site located within the coding sequence of *Amp<sup>R</sup>*.

Samples of ampicillin-sensitive,  $\beta$ -galactosidase-deficient bacteria were separately transformed with either Plasmid A or Plasmid B. The transformed bacteria were spread onto two sets of agar plates:

- **Plate 1:** Nutrient agar containing ampicillin, X-gal, and IPTG
- **Plate 2:** Nutrient agar containing X-gal and IPTG (without ampicillin)

Which row in the table correctly predicts the growth and appearance of each bacterial culture on the two plates?

|   | Plasmid A on Plate 1<br>(with ampicillin) | Plasmid A on Plate 2<br>(without ampicillin) | Plasmid B on Plate 1<br>(with ampicillin) | Plasmid B on Plate 2<br>(without ampicillin) |
|---|-------------------------------------------|----------------------------------------------|-------------------------------------------|----------------------------------------------|
| A | blue colonies                             | blue colonies                                | white colonies                            | white colonies                               |
| B | white colonies                            | white colonies                               | no growth                                 | blue colonies                                |
| C | white colonies                            | blue colonies                                | no growth                                 | white colonies                               |
| D | no growth                                 | white colonies                               | blue colonies                             | blue colonies                                |
| E | blue colonies                             | white colonies                               | no growth                                 | blue colonies                                |
| F | white colonies                            | white colonies                               | blue colonies                             | no growth                                    |
| G | no growth                                 | blue colonies                                | white colonies                            | white colonies                               |
| H | white colonies                            | blue colonies                                | blue colonies                             | no growth                                    |

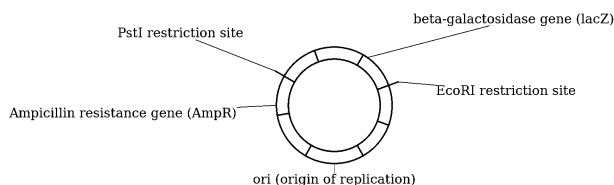


Diagram not to scale

- 26** A functional polypeptide consists of a single chain of 150 amino acids. The complete coding region of the messenger RNA (mRNA) that translates this polypeptide (excluding the stop codon) has the following base composition:

- Adenine (A): 36%
- Uracil (U): 24%
- Cytosine (C): 22%
- Guanine (G): 18%

Which row in the table correctly gives the percentage of thymine (T) in the double-stranded DNA coding region of this gene, and the total number of hydrogen bonds between the complementary base pairs in this DNA region?

|   | percentage of thymine (T) | total number of hydrogen bonds |
|---|---------------------------|--------------------------------|
| A | 24%                       | 900                            |
| B | 24%                       | 1080                           |
| C | 30%                       | 540                            |
| D | 30%                       | 1080                           |
| E | 30%                       | 2160                           |
| F | 36%                       | 900                            |
| G | 36%                       | 1080                           |
| H | 60%                       | 1080                           |

- 27** DNA sequencing was used to analyse a specific gene locus in a child and both biological parents. Paternity and maternity were genetically confirmed. The table below shows the nucleotide base present at a specific position within this locus on both chromosomes for each individual.
- | Individual | Chromosome 1 Base | Chromosome 2 Base |
|------------|-------------------|-------------------|
| Mother     | T                 | T                 |
| Father     | T                 | T                 |
| Child      | T                 | A                 |

Which statement best explains the origin of the Adenine (A) base found in the child?